

A label-free surrogate for track extrapolation in LHCb: the discrete-time physics-informed construction measured against the scheme it solves

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Abstract

Track reconstruction in LHCb calls an extrapolator, the routine that transports a five-component track state from one detector plane to another through the spectrometer magnet, millions of times a second. The standard implementation integrates the equation of motion numerically, at a cost that is accurate but neither fixed nor predictable. I build a surrogate for the field-propagation part of that routine from the discrete-time construction of Raissi, Perdikaris and Karniadakis, in which a neural network emits all internal stage states of an implicit Runge–Kutta step at once and is trained only by the requirement that those states reconstruct the starting state. No label enters the loss anywhere. The training population is harvested from an official LHCb simulation sample, 323,533 extrapolation legs of four geometrical kinds, split by particle. Every result is quoted against three fixed comparators: the same Gauss–Legendre scheme solved exactly by a root-finder, a supervised twin trained on Runge–Kutta labels under an identical protocol, and a straight line that ignores the magnet. On one frozen 5178 mm crossing of the magnet the label-free network reaches a median end-point error of $115\,\mu\text{m}$ at four layers of 200 units, three and a half orders of magnitude below the straight line and, at that size, 12 percent better than its own supervised twin. Where the discretisation is the limiting factor the network reproduces the scheme’s own error to within 0.1 and 0.6 percent at two and four stages, which is the cleanest available demonstration that the loss does what it claims. Where the network is the limiting factor the shortfall is optimisation and not capacity: over 96 converged runs spanning 12 architectures and 10 seeds, the Spearman rank correlation between final training loss and held-out endpoint error is $+0.965$. Trained on the magnet-up polarity, for which no labelled sample exists anywhere in the project, the same construction reaches the same floor, $229\,\mu\text{m}$ against $222\,\mu\text{m}$, and sits the same distance above that polarity’s own ceiling. Generalising to one network for every kind of step, with the start plane and the step length as inputs, first cost a factor 14 on the magnet crossing and left the network fifty times worse than a straight line on the short steps. Changing what the network outputs, from the absolute state to the deviation from straight-line propagation normalised by a per-leg bending integral built from the inputs alone, improved the two short leg types by factors of 92 and 190 and moved both from thirty to fifty times worse than ignoring the magnet to two to four times better than it, at $2.1\,\mu\text{m}$ on the whole test split against $16.1\,\mu\text{m}$ for a straight line. Walked along real particle paths the redesigned network is four to thirteen times more accurate than its predecessor at every chain length. The remaining constraint is the magnet crossing itself inside the shared network, $1026\,\mu\text{m}$ against a $46\,\mu\text{m}$ ceiling, which the redesign improved by only a factor 2.5 and which now dominates both the one-step and the chained error. The evidence base is 232 training runs.

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1 Introduction

LHCb is one of the four large experiments at the Large Hadron Collider at CERN, rebuilt for the current data-taking period as a fully software-triggered spectrometer [1]. Two beams of protons are brought into collision at its centre roughly thirty million times a second. Each collision sprays out charged particles, and the detector’s job is to work out which particles were produced and where they came from. It does this by placing layers of position-sensitive silicon and scintillating fibre along the beam direction, so that a particle passing through leaves a series of measured points, called hits. A **track** is the reconstructed path of one charged particle through the detector, fitted to those hits.

Between the layers near the collision point and the layers further downstream sits a large dipole magnet. Its field is roughly vertical and peaks at about 1.05 tesla around 4.7 m downstream of the collision point.¹ A charged particle passing through a magnetic field is deflected, and the amount of deflection depends on its momentum: fast particles are barely bent, slow particles are bent a great deal. The magnet is therefore the instrument that measures momentum. It is also the reason the geometry of a track is not a straight line, and the reason this paper exists.

Throughout I work in a coordinate system in which z runs along the beam direction, x is horizontal and transverse to the beam, and y is vertical. Because LHCb detects particles that go forward at small angles to the beam, every particle of interest moves monotonically in z , and it is convenient to describe a track not as a function of time but as a function of z .

A **track state** is the complete description of a track at one value of z . It is a list of five numbers,

$$S = (x, y, t_x, t_y, \text{qop}), \quad (1)$$

where x and y are the transverse position of the track at that z in millimetres; $t_x = dx/dz$ and $t_y = dy/dz$ are the local slopes, dimensionless; and the fifth component encodes the particle’s charge q , in units of the proton charge, divided by its momentum p . I use the convention of Allen, the GPU high-level trigger of LHCb [2], in which that component is stored as

$$\text{qop} = 0.299792458 \frac{q}{p[\text{GeV}]}. \quad (2)$$

Five numbers suffice, because once you know where a particle is, which way it is pointing, and its charge-to-momentum ratio, the magnetic field determines the rest of its path.

An **extrapolator** is the piece of software that takes a track state at one plane z_0 and returns the track state at another plane z_1 . That is the whole of its job, and it is used constantly: when a track fit walks along a candidate track from measurement to measurement; when the reconstruction asks whether a track found in front of the magnet and a track found behind it are

¹I write distances in millimetres throughout, because that is what the reconstruction software does, and I have never once regretted matching the code’s units rather than the reader’s intuition.

the same particle; and when a vertex fit asks where several tracks came closest to one another, which requires each of them to be transported back towards the collision point. In the high-level trigger this happens millions of times per second, on every event, inside the time budget that decides whether the event is written to disk at all.

The standard extrapolator does this by numerical integration: it steps along z in small increments, evaluating the magnetic field at each step. It is accurate and it is not cheap. A method that could take a single large step, at comparable accuracy, using a fixed and predictable amount of arithmetic, would be worth having twice over. It would be faster, and a fixed arithmetic cost is much easier to schedule on the parallel hardware the trigger runs on than a loop whose trip count depends on the track.

That is the surrogate this programme is trying to build. The restriction I work under throughout is that the surrogate replaces **field propagation only**. Real particles also scatter off the detector material and lose energy to it; those effects stay with the existing extrapolator, and none of the numbers in this paper include them.

1.1 The label-free idea

The technique I use is the discrete-time construction of Raissi, Perdikaris and Karniadakis [3], referred to throughout as *the source work*. Its distinguishing feature, and the reason I chose it over the more familiar supervised alternative, is that it needs **no labels**. The network is never shown a correct answer. It is trained only by being required to satisfy the algebra of an implicit Runge–Kutta step, which is arithmetic, together with the magnetic field, which is a measured map. Section 3.3 states the construction exactly.

If that works it is valuable for a reason beyond elegance. Producing labels for a surrogate of this kind means running the reference integrator over a large sample, and it has to be redone whenever the field map, the detector geometry or the state conventions change. A loss that never touches a label is immune to all three. The strongest version of that claim, and the one I test in Section 5.4, is that the method should work on a configuration for which no labelled sample has ever been produced.

1.2 The two questions

I set two questions at the start of the programme and stated in advance what each would mean, so that the answers could not drift to meet the results.

1. **Does it work?** By this I mean: does the label-free training converge at all; does the trained network reach the accuracy that the numerical scheme it is being trained against can itself reach; and where the network falls short of that, is the shortfall understood? A yes requires the training to be stable, the network’s error to sit on the scheme’s own error in the regime where the scheme is the limiting factor, and the remaining gap in the other regime to be attributable to something identified rather than left as a mystery.
2. **Can it be used?** By this I mean: can one trained network be pointed at an arbitrary extrapolation step, of arbitrary length, starting anywhere, and be used repeatedly along a particle’s path, at an accuracy that is better than doing nothing? A yes requires a single network to serve every kind of step, to beat the trivial baseline of ignoring the magnet on every kind of step, and to have an error that does not compound uncontrollably when the network is applied to its own output.

Question 1 is about the method. Question 2 is about the product. They are separable, and, as Section 5 shows, they came out differently.

1.3 The relation to the van der Pol study

Before any of this ran on LHCb data I rebuilt both of the source work’s constructions on a completely controlled test problem, the van der Pol oscillator, and measured them against a reference solution whose own error is negligible. That study is a companion to this one [4]. Its role here is not decorative: every design choice in the present work that could have been argued about was settled there first, on a problem small enough to measure exhaustively. Section 2 lists which findings transferred and how. The short version is that the discrete-time construction was shown, on the oscillator, to be limited by the optimiser rather than by network capacity, to be sensitive to the training seed in a way that a validation chain can diagnose, and to land within a small factor of perfect-label training at zero label cost. All three of those statements reappear in this paper, measured independently on a five-dimensional non-autonomous system with a field that is not an analytic function.

1.4 Road map

Section 2 is a related-work section: what the source work actually contains, what the van der Pol study established, and what the recent literature on the failure modes of physics-informed training says about the construction I have deliberately not used here. Section 3 derives the method from the equation of motion up, defines the loss exactly as it is implemented, and defines the three comparators, the metrics, the training protocol and the output parameterisation. Section 4 describes the data and the shared machinery. Section 5 reports each experiment in turn and then assembles the verdict. Section 6 reads that verdict against the two questions and lists the caveats honestly. Section 7 says what happens next. Appendix A carries the provenance.

2 The papers we build on

2.1 The source work, and its two constructions

Raissi, Perdikaris and Karniadakis [3] introduced physics-informed neural networks: networks trained on a loss that contains the differential equation itself, evaluated on the network’s own output, in place of or in addition to measured data. The paper contains two distinct constructions, and the distinction matters a great deal for what follows.

The **continuous-time** construction, the paper’s Section 3.1, makes the network *be* the solution. Time, and space too for a partial differential equation, go in; the solution value comes out; the residual of the equation is enforced by automatic differentiation at sampled points of the domain. It is the form most people mean by the phrase.

The **discrete-time** construction, the paper’s Section 3.2, is different in kind. The network does not represent a solution over a domain. It represents one step of an implicit Runge–Kutta scheme: given a state, it emits all the internal stage states of the step at once, and the algebra of the step supplies the supervision. The source work demonstrated it on Burgers’ equation and on the Allen–Cahn equation, in both cases taking a single very large time step with a high-stage Gauss–Legendre scheme, and reported that the construction tolerates step sizes far beyond what the same scheme would be stable at if solved conventionally. Its stated limitation is that the network is tied to one starting condition and one step: the paper trains a network for a specific initial state and a specific Δt , and says nothing about reuse. That limitation is exactly what Section 5.5 attacks, by making the starting state, the start plane and the step length inputs.

I use the discrete-time construction throughout this paper. The continuous-time one is deferred, and Section 7 says under what conditions it would be opened. The reason for deferring it is not taste. It has no fixed arithmetic cost per step, which is the property the trigger wants; it imposes a much weaker constraint per training point; and, as the next two subsections record, it

has a documented failure mode on long horizons that the discrete-time construction structurally cannot have, because a discrete-time network is never asked to represent a trajectory over an interval at all.

2.2 What the van der Pol study established

The companion study [4] rebuilt both constructions on the van der Pol oscillator under identical optimiser discipline. Four of its findings shaped this programme directly.

The capacity floor is an optimisation floor. A 135-run architecture sweep on the discrete-time construction found an endpoint error floor that larger networks did not remove, and identified it by measuring the rank correlation between the final training loss and the held-out one-step error: +0.974. Every architecture and every seed lay on one loss-error band. The conclusion was that a wider network helps only because the optimiser gets further down on it, not because the narrower network could not represent the answer. Section 5.3 repeats that measurement on the magnet crossing and gets +0.965, which is the same statement about a very different system.

The seed is a real degree of freedom, and the chain selects it. Long-horizon quality on the oscillator was decided by the training seed, was measurable on a validation chain, and transferred to held-out starts. The protocol that came out of it, train ten seeds, chain each to the target horizon on validation data, keep the best, is the protocol Section 5.5 uses on real particle paths.

Supervision is the right control, not the right method. The oscillator study ran a matched data-supervised twin: same network, same initialisation, same optimiser, same budget, physics loss swapped for mean squared error against the reference. The equation-trained network landed within a factor 2.1 of perfect-label training at zero reference-data cost. That twin is reproduced here for every experiment, and Section 5.3 reports the one regime where the label-free loss overtakes it.

Chaining a reusable one-step map beats one large step. On the oscillator, generalising the discrete-time network so that its input is the starting state itself, and then applying it repeatedly, beat the source work’s single-large-step mode by a factor 10 at the largest feasible step. That is the design used here from the outset: the network is a one-step map, and Section 5.5 walks it.

2.3 The stability line, and why it argues for the discrete-time column

The continuous-time construction has a failure mode that is now well characterised, and understanding it is the main reason the parked column stays parked. I summarise the three papers because they bear directly on what would happen if the LHCb work opened that line.

Rohrhofer, Posch, Gößnitzer and Geiger [5] observe that if the network output is a constant equal to a fixed point y^* of the flow, so that $f(y^*) = 0$, then its time derivative is zero and the residual is exactly zero at every collocation point, wherever those points sit. Fixed points are therefore global minima of the residual term, with basins that grow with the horizon, and only the initial-condition term prefers the true solution. They measure success falling monotonically with horizon and with proximity of the start to a fixed point, and report that larger networks repair nothing, concluding that the difficulty is bound to optimisation complexity rather than to expressive power. They propose no remedy.

Wang, Koohy, Lu and Perdikaris [6] sharpen this into a statement about the whole family of spurious minimisers. For any fixed collocation set there exists a candidate that follows the true solution near every collocation point before some time t_0 , follows a different solution of the same equation after it, and hides the hand-over in the gaps between collocation points, so that its empirical residual is *exactly* zero. The fixed point is only the cheapest member of that family. Their remedy is pseudo-time stepping with collocation resampling, which penalises

change from the previous iterate and thereby amplifies the hidden transition layer. Babic, Rohrhofer and Geiger [7] instead propose a regulariser that penalises a candidate for sitting still inside a locally unstable region, a product of the positive real parts of the Jacobian eigenvalues at the network’s output and a Gaussian in the network’s own speed, inert on any moving candidate by construction, and demonstrate it on van der Pol to about two periods.

The van der Pol study measured all three. The finding that matters here is the ordering: the regulariser turns parking at the fixed point into a different member of the same family, a flow-following splice, and only pseudo-time stepping prices the whole family rather than one member of it. Across 240 converged runs, pseudo-time stepping was the only intervention that worked past two periods.

None of this applies to the discrete-time construction as such. A discrete-time network is asked for the stage states of one step, and the reconstruction residual of Section 3.3 is not zero for a constant output: the stage equations tie every output to the known starting state, which enters the loss as data. There is no interval over which a hand-over can be hidden and no free phase. That structural difference, rather than any measured comparison on LHCb data, is why the discrete-time column was opened first.

2.4 Gauss–Legendre collocation

The numerical scheme underneath everything here is the Gauss–Legendre family of implicit Runge–Kutta methods, whose classical theory is set out by Hairer and Wanner [8]. Two of its properties are load-bearing. A q -stage member has formal order $2q$, the maximum attainable by any q -stage Runge–Kutta method, which is what makes it the right family for a method that intends to take very few, very large steps. And the family is equivalent to polynomial collocation: the q -stage scheme is exactly the polynomial of degree q through the step that satisfies the differential equation at the q node positions. Section 3.2 uses the second property as the mental picture throughout.

3 Theory

This section builds the method from the equation of motion up. I write down the differential equation that governs a track in z and say where each term comes from; explain what makes a Runge–Kutta method implicit and why the Gauss–Legendre family is the natural choice; describe the construction that turns that scheme into a training objective with no labels; define the loss and its scales; define the three comparators and the metrics; state the training protocol and the fiducial requirement; and finally define the output parameterisation that Section 5.6 tests.

3.1 The equation of motion in z

A charged particle in a magnetic field $\mathbf{B}$ feels the Lorentz force $q\mathbf{v} \times \mathbf{B}$. Rewriting Newton’s second law with z rather than time as the independent variable, and expressing the result in the state variables of Eq. (1), gives the system that `_shared/reference.py` implements:

$$\frac{dx}{dz} = t_x, \quad \frac{dy}{dz} = t_y, \quad (3)$$

$$\frac{dt_x}{dz} = \kappa \text{qop } N [t_x t_y B_x - (1 + t_x^2) B_y + t_y B_z], \quad (4)$$

$$\frac{dt_y}{dz} = \kappa \text{qop } N [(1 + t_y^2) B_x - t_x t_y B_y - t_x B_z], \quad (5)$$

$$\frac{d\text{qop}}{dz} = 0, \quad (6)$$

where $\kappa = 10^{-3}$ is the unit conversion constant that goes with the Allen qop convention of Eq. (2); $N = \sqrt{1 + t_x^2 + t_y^2}$ is the ratio of path length to z length, which appears because the force acts along the path while we differentiate along z ; and B_x, B_y, B_z are the three components of the field in tesla, evaluated at the point (x, y, z) the particle currently occupies.

Three properties of this system matter for everything that follows.

- **It is non-autonomous.** The right-hand side depends on z explicitly, through the field. This is unlike the textbook test problems on which the technique was originally demonstrated, where the right-hand side depends only on the state.
- **qop is conserved,** by Eq. (6). A magnetic field does no work, so restricting the surrogate to field propagation makes the momentum an input that is carried through unchanged. Only four components are ever predicted.
- **The field is not an analytic function.** It is a measured map, stored on a grid, and Section 3.6 shows that this sets the ceiling on the whole exercise.

Analogy. Equations (3) to (5) say the same thing as a description of a car on a road: the position changes at a rate given by the current heading, and the heading changes at a rate given by how hard the steering wheel is turned. The magnetic field is the steering wheel, and qop is how strongly the car responds to it. A heavy, fast car, meaning high momentum and small qop, barely turns; a light, slow one turns sharply.

Worked micro-example. Take a particle with $p = 10$ GeV and charge $+1$, so $\text{qop} = 0.0300$ by Eq. (2). Take it travelling nearly along the beam, $t_x = t_y = 0$, so $N = 1$, through a field of $B_y = -1.0$ T with $B_x = B_z = 0$. Then Eq. (4) gives $dt_x/dz = \kappa \text{qop} (-1) B_y = 10^{-3} \times 0.0300 \times 1.0 = 3.0 \times 10^{-5}$ per millimetre. Over the 5178 mm crossing of the magnet studied here, and treating the field as constant for the estimate, the slope would grow to about 0.155 and the particle would move sideways by about $0.155 \times 5178/2 \approx 400$ mm. The measured median difference between the true path and a straight line on that leg is 520 mm, the same order. So the quantity the surrogate has to get right is a half-metre correction, and getting it right to $100 \mu\text{m}$ is getting it right to one part in five thousand.

3.2 Implicit Runge–Kutta methods as collocation

Given an initial state $S(z_0)$ and the derivative function $f(S, z)$ of Section 3.1, a numerical scheme advances the solution by a step Δz . The general Runge–Kutta method with q stages does so by evaluating f at q intermediate points and combining the results:

$$S_j = S_0 + \Delta z \sum_{k=1}^q A_{jk} f(S_k, z_0 + c_k \Delta z), \quad j = 1, \dots, q, \quad (7)$$

$$S_{\text{end}} = S_0 + \Delta z \sum_{k=1}^q b_k f(S_k, z_0 + c_k \Delta z), \quad (8)$$

where S_0 is the state at the start of the step; the S_j are the q **stage states**, the method's internal guesses at the solution at intermediate planes; $c_k \in [0, 1]$ are the **nodes**, the fractional positions of those planes within the step; A_{jk} is the $q \times q$ **stage matrix**; and the b_k are the q **weights** that combine the stages into the final answer. Together (c, A, b) are the tableau.

If A is strictly lower triangular the scheme is **explicit**: stage 1 depends on nothing, stage 2 on stage 1, and so on, so the stages can be computed one after another. The classical fourth-order method used as the reference throughout this work is of that kind. If A is full the scheme is **implicit**: every stage depends on every other stage, including itself, and Eq. (7) is a coupled nonlinear system that has to be solved.

The **Gauss–Legendre** family is the implicit family obtained by choosing the nodes c_k to be the roots of the degree- q Legendre polynomial mapped onto $[0, 1]$, and then choosing A and b so that the scheme integrates polynomials exactly to the highest possible degree. The result has formal order $2q$ [8]. It is also a **collocation** method: it is exactly equivalent to fitting a polynomial of degree q through the step and requiring that the polynomial satisfy the differential equation exactly at the q node positions. The stage states are the values of that polynomial at the nodes.

Analogy. Explicit stepping is like walking across a room in small steps with your eyes on your feet: each step follows from the last. Collocation is like drawing a single smooth curve from one side of the room to the other and then checking, at eight chosen points along it, that the curve is heading in the direction the physics says it should be heading at that point. If it passes all eight checks, and the curve is smooth, it is very likely right everywhere.

Worked micro-example: the $q = 2$ tableau. The nodes are the roots of the shifted Legendre polynomial of degree two, $c = \left(\frac{1}{2} - \frac{\sqrt{3}}{6}, \frac{1}{2} + \frac{\sqrt{3}}{6}\right) \approx (0.2113, 0.7887)$, the weights are $b = (\frac{1}{2}, \frac{1}{2})$, and

$$A = \begin{pmatrix} 1/4 & 1/4 - \sqrt{3}/6 \\ 1/4 + \sqrt{3}/6 & 1/4 \end{pmatrix} \approx \begin{pmatrix} 0.2500 & -0.0387 \\ 0.5387 & 0.2500 \end{pmatrix}. \quad (9)$$

The entry $A_{12} = -0.0387$ is nonzero and sits above the diagonal, so the first stage depends on the second: there is no order in which these can be evaluated one at a time. That is exactly what implicitness means. The closed-form tableaux for $q = 1, 2, 3$ are hard-coded in `_shared/irk.py` and used to check the general construction, which builds the tableau for any q from the Legendre roots.

3.3 The discrete-time construction: training with no labels

Ordinarily Eqs. (7) and (8) are a problem to be solved: given S_0 , find the q stage states and the endpoint that satisfy the $q + 1$ equations. Solving them requires a root-finder, which is iterative and whose cost is not known in advance.

The discrete-time construction of the source work [3] inverts the arithmetic. A network is asked to **output** all q stage states and the endpoint at once, given only the starting state:

$$\mathcal{N}_\theta : S_0 \mapsto (S_1, S_2, \dots, S_q, S_{\text{end}}), \quad (10)$$

where θ are the network’s weights. Its outputs are then substituted into the scheme’s equations, read backwards. Each of the $q + 1$ equations can be rearranged to express the *starting* state in terms of a proposed stage or endpoint:

$$S_0^{(j)} = S_j - \Delta z \sum_k A_{jk} f(S_k, z_0 + c_k \Delta z), \quad (11)$$

$$S_0^{(\text{end})} = S_{\text{end}} - \Delta z \sum_k b_k f(S_k, z_0 + c_k \Delta z). \quad (12)$$

Each of the $q + 1$ outputs therefore produces its own **reconstruction** of the starting state. If the network’s proposal is a genuine solution of the scheme, all $q + 1$ reconstructions equal the true S_0 , which is known because it is the input. If it is not, they do not. The discrepancies are the reconstruction residuals, and they are the entire training signal:

$$r_{n,d}^{(j)} = \frac{S_{0,n,d}^{(j)} - S_{0,n,d}}{\sigma_d}, \quad \mathcal{L}_{\text{phys}}(\theta) = \frac{1}{(q+1)d_{\text{max}}N} \sum_{n,j,d} \left(r_{n,d}^{(j)}\right)^2, \quad (13)$$

where n runs over the N training states, j over the $q + 1$ outputs, d over the $d_{\text{max}} = 4$ predicted components, and σ_d is the per-component scale of Section 3.4. This is the mean

squared normalised reconstruction residual, and it is implemented in `_shared/model.py`, function `physics_loss`, in exactly this form.

Nothing in Eq. (13) is a label. No correct answer ever enters it. What enters it is the starting state, which is the input; the tableau, which is arithmetic; and the magnetic field, which is a measured map. The field enters through f , evaluated at the positions the network itself proposes, which is why the field must be available in a differentiable form: the gradient of the loss with respect to the weights runs through the field lookup.

Analogy. Instead of solving a maze by walking it, you hand someone a proposed route and ask them to walk it backwards from every waypoint. If every backwards walk lands on the entrance, the route is right. Nobody ever had to be told the correct route.

Worked micro-example, and the cost argument. Take $q = 8$ and one training state. The network emits $9 \times 4 = 36$ numbers. The rates f are evaluated once at each of the eight stage states, requiring eight field lookups. The eight stage reconstructions of Eq. (11) and the one endpoint reconstruction of Eq. (12) are formed by two small matrix products and compared against the four known components of the input. That is 36 residual numbers from one forward pass and eight field lookups, with no iteration and no branch. Solving the same eight-stage system directly with a root-finder took, on the cross-magnet leg, a median of 44 residual evaluations, each of which is itself eight field lookups plus the root-finder’s internal Jacobian work. The arithmetic advantage of the trained network is not that it is approximate; it is that its cost is fixed and known in advance.

3.4 The scales in the loss

The state components have very different units and sizes: x is hundreds of millimetres, t_x a few tenths and dimensionless. Squaring and summing them directly would let x dominate the loss entirely. Each residual component is therefore divided by a fixed scale σ_d before squaring, making Eq. (13) dimensionless.

The scales are measured from the training population itself, and are label-free: they are the spread of each component over the training states. Measured on the full training split across all leg types they are $\sigma_x = 655.6$ mm, $\sigma_y = 360.2$ mm, $\sigma_{t_x} = 0.2592$, $\sigma_{t_y} = 0.1152$ and $\sigma_{qop} = 0.1190$. Each experiment recomputes them on its own dataset; for the frozen cross-magnet leg they are $\sigma_x = 176.5$ mm and $\sigma_y = 182.7$ mm on the input side, with wider output-side values of 396 mm and 417 mm reflecting the spread the states acquire after crossing the magnet.

One scale per component for the whole population is a deliberate choice, inherited unchanged from the verified baseline. It turns out in Section 5.5 to be the leading suspect for the general network’s poor performance on short steps, and Section 3.10 states the replacement.

3.5 The three reference points

A number in micrometres means nothing on its own. Every result in this paper is quoted against three fixed comparators.

1. **The exact scheme.** The same q -stage Gauss–Legendre equations, on the same states, solved directly by a root-finder in double precision, with no network anywhere. This is the **ceiling**: the best that any network trained on those equations could possibly do, because it is what the equations themselves say. Convergence is declared at a residual below 10^{-8} in units of millimetres and milliradians.
2. **The data twin.** The identical network, identical initialisation, identical optimiser and identical budget, with the physics loss of Eq. (13) swapped for a plain mean squared error against reference stage and endpoint states computed by fourth-order Runge–Kutta. This is the supervised control. It answers the question of what the label-free loss cost us, and it is the same control the source work used.

3. **The straight line.** What you get by ignoring the magnet entirely and continuing along the incoming slopes, $x_0 + t_x \Delta z$ and $y_0 + t_y \Delta z$. This is the trivial baseline, and any surrogate that does not beat it is doing nothing useful.

3.6 Why the ceiling floors at about $23\mu\text{m}$

The formal order of the q -stage Gauss–Legendre scheme is $2q$, so one might expect the exact scheme’s error to fall as $(\Delta z)^{2q}$ and keep falling as q increases. It does not. Measured on the frozen cross-magnet leg, the ceiling falls from $5401\mu\text{m}$ at $q = 4$ to $23\mu\text{m}$ at $q = 8$, and then to only $20\mu\text{m}$ at $q = 16$. It stops.

The reason is the field map. The canonical LHCb field map v8r1 is a measured table on a regular grid of $81 \times 81 \times 146$ points spaced 100 mm apart, spanning -4000 to $+4000$ mm in x and y and -500 to $+14000$ mm in z . The field at an arbitrary point is obtained by **trilinear interpolation** between the eight surrounding grid values. A trilinearly interpolated function is continuous but its derivative is not: at every cell boundary the gradient jumps. In the standard classification the interpolant is C^0 and not C^1 .

High-order integration schemes derive their order from Taylor expansions of the right-hand side, and those expansions require the right-hand side to be differentiable. It is not. A 5178 mm step across the magnet crosses about 52 cell boundaries in z alone, and many more in x as the track bends. Beyond the point where the stages are dense enough to resolve the overall shape of the field, additional order buys nothing, because there is no additional smoothness for it to exploit. The scheme converges to a floor set by the kinks in the interpolant, and on this leg that floor is about $23\mu\text{m}$ against a reference that resolves those same kinks finely.

Analogy. Fitting a high-degree polynomial through a curve made of straight line segments: past a certain degree, the extra flexibility is spent chasing the corners, and the fit stops improving.

Worked micro-example. On the same test states, at $q = 8$ the exact scheme’s median endpoint error is $22.544\mu\text{m}$ and its 95th percentile is $1610\mu\text{m}$; at $q = 16$ the median is $19.5\mu\text{m}$ and the 95th percentile is $325\mu\text{m}$. The median moved by 13 percent while the number of unknowns doubled. The tail improved much more than the median, which is consistent with the tail being dominated by soft, hard-bending tracks whose paths sample the map most aggressively.

This $23\mu\text{m}$ figure is the single most important reference number in the paper. It is not a property of any network. It is what the mathematics being solved is worth on this field map.

3.7 Metrics

Every error quoted in this paper is defined identically, in `_shared/evaluate.py`.

- **Endpoint error:** for each test state, the larger of $|\Delta x|$ and $|\Delta y|$ between the prediction and the double-precision fourth-order Runge–Kutta reference at the end plane, converted to micrometres. Quoted as the median over the test states, with the 95th percentile where the tail matters.
- **Stage error:** the same quantity taken over all q interior stage states at once.
- **Slope error:** the larger of $|\Delta t_x|$ and $|\Delta t_y|$ at the endpoint, in milliradians.
- ρ : a dimensionless relative error, $\rho = \min(1, \|S_{\text{ref}} - S_{\text{pred}}\|^2 / \|S_{\text{ref}}\|^2)$, Euclidean over the four dynamic components. This is the capped measure agreed for the van der Pol study [4], applied here to the same four components.
- **Straight-line error:** the endpoint measure applied to the straight-line baseline of Section 3.5.

Where a result is quoted over several random seeds, I give the median over the seeds of the per-seed median over test states, with the range across seeds in brackets. A single seed is not a measurement, and Section 5.3 quantifies how much it can differ.

3.8 The training protocol: stall and confirm

Every run in this paper followed the same numbered protocol, unchanged from the verified July baseline.

1. Build the dataset. Draw training states from the official sample’s train split, transport each one exactly onto the leg’s start plane if it is within 60 mm of it, compute the label-free scales of Section 3.4 from the resulting population, and apply the fiducial requirement of step 2.
2. **Apply the fiducial requirement.** Discard any state whose reference trajectory leaves the field map. This removes about one percent of states on the magnet-down polarity. Section 3.9 says why.
3. Initialise a fully connected network with hyperbolic-tangent activations, in double precision throughout, with five inputs, two more for the general-leg experiments carrying the start plane and the step length, and $4(q + 1)$ outputs.
4. Train with full-batch L-BFGS, 200 iterations per restart, strong Wolfe line search, history 120, single-threaded on one CPU core.
5. **Stall.** Restart the optimiser repeatedly until two consecutive restarts each improve the training loss by less than one percent.
6. **Confirm.** Start a completely fresh optimiser, with its curvature history discarded, from that point. The run counts as converged only if it re-stalls within two restarts without moving the train, validation or test endpoint medians by more than one percent. Anything else is reported as unconverged and never pooled into a median.
7. Score on the held-out test split, which no run ever trained on and which contains no particle that appears in training.

Steps 5 and 6 together are what I call stall-and-confirm. They exist because of a lesson from July: the first attempt at this experiment used a fixed budget of six optimiser restarts, and every run was still improving by 20 to 35 percent per restart when the budget ran out. Those results were budget-limited rather than converged, and they gave a factor-two gap between the physics loss and the supervised twin that turned out not to be real.²

3.9 The fiducial requirement

The fiducial requirement has a similar history and is worth stating carefully, because it is the main portability lesson of the whole programme.

On the first run against the official sample the physics loss appeared much worse than the twin, 403 to 595 μm against 175 to 213 μm . The cause was that 21 of 2000 training states, one percent, are soft tracks of 1.2 to 1.5 GeV that bend so hard they leave the field map entirely, past $|x| = 4$ m, which is outside the LHCb acceptance as well as outside the map. Those 21 states carried **half of the entire physics loss**, and the top one percent of states carried 68 to 78 percent of it.

²A fixed budget is a comfortable thing to report and a dangerous thing to believe. It took two independent experiments to convince me that a number which is still moving is not a number.

The asymmetry between the two losses is structural. The physics loss evaluates the field at the positions the network proposes, so a trajectory outside the map is being asked to satisfy equations built on a clamped, meaningless field, producing a residual the optimiser cannot reduce and which dominates the gradient. The supervised twin never touches the field: it fits its labels, which were computed with the same clamped field, and absorbs the pathology silently.

Requiring the reference trajectory to stay inside the region where the equations are defined at all removes the pathology. With the requirement in place the physics runs stopped stalling at a plateau, taking 147 to 152 restarts instead of 57 to 72, and the training loss fell from about 1.1×10^{-4} to about 8.3×10^{-6} , the same range as the twin. The cut removes 21 train, 20 validation and 15 test states out of 2000/2082/2033 on the frozen leg.

3.10 The straight-line-residual parameterisation

Everything above leaves one design choice implicit: the network of Eq. (10) emits the *absolute* state at each stage and at the endpoint, in units of a single population-wide scale per component. That is the source work’s own choice, and on a single frozen leg it is unobjectionable because the population is one geometry. On a population spanning steps from 70 mm to 8.6 m it is not, and Section 5.5 measures the damage.

The replacement, implemented in `General_leg_network/residual_model.py`, changes exactly one thing: what the network is asked to output. For a start state S_0 at z_0 , a step Δz , the Gauss–Legendre nodes $z_j = z_0 + c_j \Delta z$ and the endpoint $z_0 + \Delta z$,

$$\text{output}_j = \text{straight}_j + \text{scale} \odot \text{net}_j, \quad \text{straight}_j = (x_0 + t_x(z_j - z_0), y_0 + t_y(z_j - z_0), t_x, t_y), \quad (14)$$

where $\odot$ is the componentwise product and net_j is the raw network output, now a pure number rather than a length. The network predicts the deviation from a straight line, which is the magnet’s whole contribution and nothing else.

The scale, and why it may not use a label. The network is never shown a label, so the scale may only be built from the inputs. To first order in the field, Eqs. (4) and (5) give

$$|\Delta t_x| \approx \kappa |\text{qop}| I_B, \quad |\Delta x| \approx \frac{1}{2} \kappa |\text{qop}| I_B |\Delta z|, \quad (15)$$

the first dimensionless and the second in millimetres, being that slope ramped over the step. Here

$$I_B = \int_{z_0}^{z_0 + \Delta z} |\mathbf{B}(x_s(z), y_s(z), z)| dz \quad [\text{T mm}] \quad (16)$$

is the **bending integral**, the field magnitude integrated along the *straight line* $x_s(z) = x_0 + t_x(z - z_0)$, $y_s(z) = y_0 + t_y(z - z_0)$, evaluated with a 16-point midpoint rule. Every symbol on the right of Eqs. (15) and (16) is an input. Floors of 10^{-3} mm and 10^{-6} keep the scale away from zero for a straight track in a field-free region.

Equation (15) is a first-order estimate of the answer, which is exactly what a scale should be. It does not have to be right. It has to be right to within a factor of a few on every leg type, so that what the network learns is an $O(1)$ correction to it, whatever the geometry. That is what makes every leg $O(1)$: a 70 mm sensor hop and a 5.2 m magnet crossing differ by five orders of magnitude in the physical correction, and by the same five orders of magnitude in $I_B |\Delta z|$, so the ratio the network must emit is of order one in both cases.

Two properties recommend the design beyond conditioning. The quantity being learned is the physics, the bending, rather than the geometry, meaning where the plane happens to be. And it is zero when there is no field, so the trivial baseline of Section 3.5 is built into the parameterisation and the network cannot be worse than a straight line by construction unless it is actively wrong.

The check that had to pass first. Before any farm time was spent, `prepare_residual.py` formed the ratio the network will actually have to emit, that is (reference minus straight line) divided by the scale, on the test split, and histogrammed it by leg type. This is the one place a label appears in the residual arm, and it is a diagnostic, never an input to the scale. Two node profiles were considered: *flat*, one scale for all $q + 1$ outputs of a sample, and *poly*, output j scaled by c_j^2 for positions and c_j for slopes. The rule was fixed before looking: take flat unless its endpoint medians fall outside 0.1 to 10 on some leg while poly’s do not. Both passed, with medians 1.09, 1.03 and 0.95 on the three leg types, so the rule selected flat. The same label-free expression lands within ten percent of the truth on a 341 mm backward vertex fetch, a 5.2 m magnet crossing and a 70 mm sensor hop.

What else changed, and what did not. Nothing else. The dataset is byte for byte the same population plus three scale arrays per split; the physics loss of Eq. (13) is computed from the reconstructed absolute states and needed no change; the optimiser protocol, scoring, splits, momentum bands and ceilings are all Section 3.8’s. The one deliberate consequential change is that the data twin’s normalisation follows: the supervised loss divides by the per-sample residual scale rather than by the population output scale, because otherwise the twin would be trained on an objective the physics arm is not.

4 Data and machinery

4.1 The sample

The training population is derived from official LHCb simulation, not from anything I generated. This was a deliberate change of policy on 21 July 2026: small configuration errors in home-grown event generation can cause large and silent problems, and a centrally produced sample carries its own validated conditions.

The sample is the LHCb TestFileDB entry `expected_2024_minbias_xdigi`, production 00212966, in XDIGI format, meaning digitised detector banks plus the packed simulation truth, with detector description `dddb-20231017` and conditions `sim-20231017-vc-mu100` for 2024 data-taking. Five of the entry’s eight files, 19 GB, are mirrored on the local CVMFS filesystem, and 200 events were read from them. No grid access was needed.

Correction, 6 September 2026: the sample is magnet-up. The conditions tag `sim-20231017-vc-mu100` is the **magnet-up** configuration. This was established while building the magnet-to-magnet data set of the next block, by propagating each particle’s forward cross-magnet leg from its last upstream-tracker plane and comparing the result with that particle’s *own* first scintillating-fibre state, which the sample supplies for free. Over all 15,257 selected particles the leg propagated with `field.v8r1.up.bin` lands a median 1.73 mm from the real state (14.8 mm at 1–2 GeV, 0.28 mm at 25–200 GeV), with the sign of the bend correct for 100.0 percent of particles; propagated with `field.v8r1.down.bin` it lands 883 mm away (3973 mm and 169 mm in the same two bands) with the sign wrong for 100 percent, and 2 percent of legs leave the field map altogether. The two maps are exact negatives, so polarity changes only the sign of the bend.

Every label and score in this paper was computed with the magnet-down map: the Y column of the training set, the frozen-leg experiments of Sections 5.1 to 5.3, and the general-leg and residual experiments of Sections 5.5 and 5.6. Four consequences follow, and no more.

1. **Every method conclusion stands.** Each experiment compared a network against the same equations and the same engine: scheme-limited against network-limited, the optimisation-limited floor, the residual redesign, and chaining are all internal comparisons and are untouched.
2. **Any statement that those labels are where the simulated particle went is wrong,** and the material-floor gate G2 of Section 4.3 must be re-read on the magnet-up map.
3. **The magnet-up experiment of Section 5.4 is not “a polarity with no labels” but the sample’s true polarity,** and it is the only experiment here scored against the physically right field. Its result is unchanged — the same floor, 229 against $222\ \mu\text{m}$ — and its framing is not.
4. The magnet-to-magnet data set, the fine reference and the tables now in progress are built on the magnet-up map.

Source: `R/Magnet_tracks_dataset/results/polarity_check.json` and `R/Magnet_tracks_dataset/README.md`.

4.2 The harvest, and the four leg types

1. Dump the simulation truth from the XDIGI stream: every simulated particle, every vertex, and every hit each particle deposited in a tracker sensor.
2. For each charged particle, turn each tracker hit into a state: the position where it crossed the sensor plane, the local slopes there, and the momentum from truth.
3. Form **legs**, meaning pairs of planes to be extrapolated between, from each particle’s own geometry. Four kinds are built, listed in Table 1.
4. Compute the label for each leg by integrating Eqs. (3) to (6) with fourth-order Runge–Kutta in double precision with 5 mm steps, using the canonical field map at magnet-down polarity.³ The labels are used to score every experiment and to train the data twin; the physics loss never touches them.
5. Split by **particle**, 80/10/10, with seed 20260718, so that no particle contributes legs to more than one split. This matters: legs from one particle are highly correlated, and splitting by leg would leak.

4.3 The integrity gates on the labels

Four gates were run on the labels and are recorded in the dataset metadata.

³The sample itself is magnet-up; see the correction box in Section 4.1. The labels are therefore field-only propagations of the right start states through the wrong polarity, which is a statement about what the labels *are* and not about any comparison made with them.

Table 1: The four leg types and the row counts of the harvested set. A leg is a pair of planes to be extrapolated between. Type D is built and labelled but never trained on; it exists as an out-of-training test in Section 5.5. Counts are over the whole set, before any experiment’s own subsampling or fiducial cut.

leg	what it is	typical step	rows
A	vertex fetch: first vertex-detector state back to its vertex	341 mm, backward	49,014
B	cross-magnet: last plane before the magnet to the first after, both directions	5175 mm	41,398
C	plane-to-plane: consecutive sensor crossings, forward, at most three per particle	70 mm	205,887
D	downstream to vertex: first post-magnet state back to the vertex	−8.6 m	27,234
total, split by particle into 258,857 train / 32,389 validation / 32,287 test			323,533

Table 2: Integrity gates on the reference labels, from `train_official_v2.meta.json`. G2 is the interesting one and is discussed in the text.

gate	what it checks	result
G1	closure	propagate forwards then backwards; the state must return to where it started
G2	truth	the label compared with where the particle’s own next simulated hit actually is
G3	step	repeat the integration with 1 mm steps instead of 5 mm
G4	conservation	qop must pass through unchanged

Gate G2 defines the scope of the whole project. The label disagrees with the particle’s real next hit by an amount that grows as momentum falls, following the characteristic $1/p$ behaviour of multiple scattering off detector material. That residual is not an error in the labels. It is the material effect, which is deliberately outside the surrogate’s remit and stays with the existing extrapolator.⁴

4.4 The shared package and its three gates

All experiment folders import one package, `R/_shared`, so that a difference between two experiments is a difference in the experiment and never a drifted copy of the physics or the optimiser. The package contains the field-map loader, the tableau construction, the reference integrator, the differentiable field, the network, the losses, the dataset builders, the scorers and the training driver. Three gates protect it, and all three pass.

1. **Vendoring parity.** The field-map loader was copied into this repository from its original location, and the copy is required to return bit-identical values to the original on 200,000 random points inside the map. Measured difference: exactly 0.0 tesla, and the map file’s md5 matches.
2. **Differentiable-field parity.** The physics loss needs the field as a function it can differentiate through, so there is a second implementation in PyTorch. It is required to agree with the numerical original to 10^{-12} T. Measured: 1.3×10^{-15} T on both polarities.
3. **Baseline reproduction.** The generalised code, with its generalisations switched off, must reproduce the verified July baseline bit for bit: the same dataset of 1979/2062/2018 states

⁴G2 as tabulated was measured against labels computed with the magnet-down map, so on the cross-magnet leg it is not a material-only residual and must be re-read on the magnet-up map. See the correction box in Section 4.1.

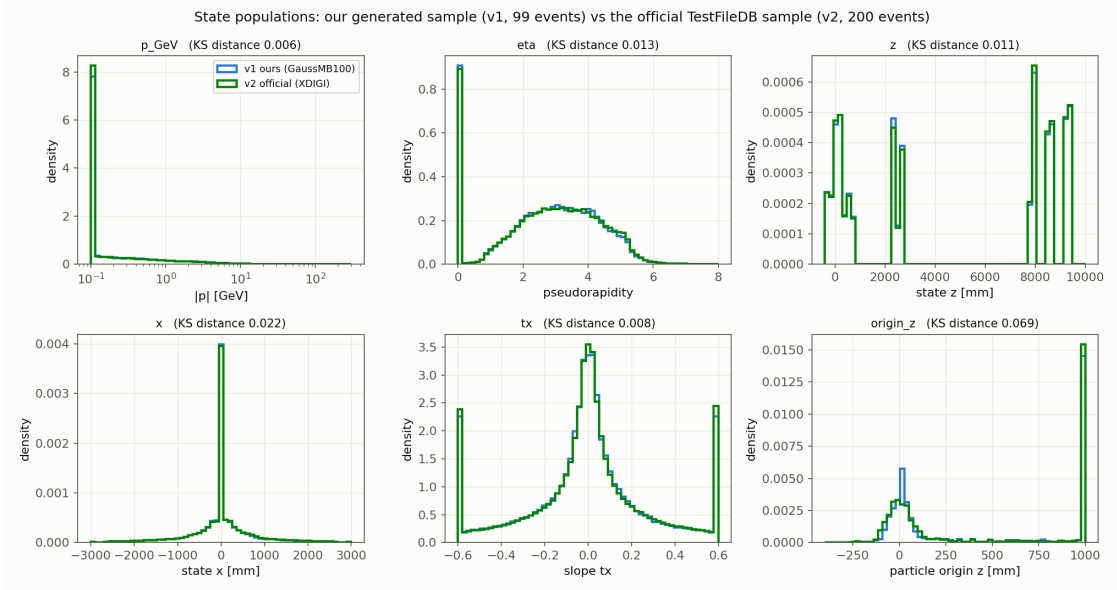


Figure 1: The population the surrogate is trained on, compared between the sample generated in-house in July (v1, 99 events) and the official TestFileDB sample that replaced it (v2, 200 events). The six panels show momentum, pseudorapidity, the z position of the harvested states, the transverse position x , the slope t_x , and the z of the particle’s production point, each as a normalised density with the Kolmogorov–Smirnov distance between the two samples printed in the panel title. The distributions agree closely, with distances 0.006 to 0.069, which is the check that switching to the official sample changed the provenance of the population and not its physics. The structure in the z panel is the detector itself: the three clusters are the vertex detector around $z = 0$, the upstream tracker near 2400 mm, and the scintillating-fibre tracker beyond 7800 mm. The peak at 1000 mm in the origin panel is particles produced in material rather than at the collision point; those are kept in the sample and flagged, not hidden.

with identical arrays, the same initial parameters, the same losses and gradients, and the first optimiser restart landing on the same number to the last bit.

4.5 Compute, and the single-thread caveat

All training ran on the Nikhef HTCondor farm, one job per combination of architecture, mode and seed, single-threaded and in double precision. Multithreaded PyTorch spin-waits on these nodes and is about 68 times slower, so every job sets one thread before importing the library. A consequence worth stating plainly is that runs from the September campaign are not bit-comparable with the July runs, which used four threads: a different reduction order changes the last bits, and 200 line-searched iterations amplify that into the fourth significant digit. Comparisons across that boundary are made at the level of seed ranges, never digit for digit.

5 Results

This section reports the July baseline the campaign starts from, then each experiment in turn, then assembles the verdict.

5.1 The July baseline

By the end of July three things were fixed and have not been revisited since.

The leg. The experiments needed one concrete extrapolation step to work on. I took the modal plane pair of the 11,046 forward cross-magnet legs in the training set: $z_0 = 2648.2$ mm, the last plane of the upstream tracker, to $z_1 = 7826.0$ mm, the first plane of the scintillating-fibre

Table 3: The July baseline on the frozen cross-magnet leg, 4×50 network, $q = 8$, three seeds. Test-split median endpoint error in micrometres, with the individual seeds in brackets. The two losses are indistinguishable within the seed spread. Source: `One_step_network_v2/results/summary.csv`.

	endpoint error, μm	
exact scheme, $q = 8$ (the ceiling)		23
network, label-free physics loss	183	[177, 183, 235]
network, supervised data twin	192	[162, 192, 208]
straight line (ignore the magnet)		520,444

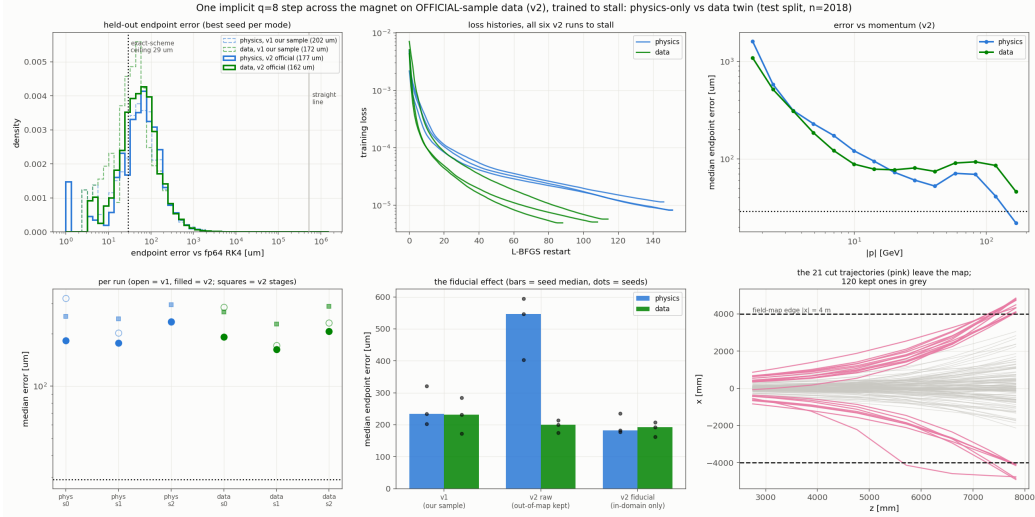


Figure 2: The July baseline on the frozen magnet crossing. The panels show the distribution of test-split endpoint errors for the three label-free seeds and the three supervised seeds, the training histories per optimiser restart with the confirmation pass marked, and the comparison against the in-house sample the official one replaced. The point of the figure is the overlap of the two loss families: with the fiducial requirement of Section 3.9 in place, the label-free loss and the supervised twin are not distinguishable at three seeds on this leg.

tracker. The step is $\Delta z = 5177.8\text{mm}$ and it crosses the whole magnet. I call it the frozen leg. It is the hardest of the routine steps and the one where a surrogate would be most valuable.

The number of stages, $q = 8$. Chosen from the exact-scheme measurement of Section 3.6: at $q = 8$ the scheme’s own error on this leg has fallen to the field map’s floor, and going further buys nothing. Section 5.2 tests that choice properly.

The first converged label-free result. On the official sample, with the fiducial requirement in place and trained to a confirmed stall, a network of four layers of 50 units with $q = 8$ reached the numbers of Table 3.

That is the result the September campaign set out to interrogate.

5.2 How many stages the magnet crossing needs

Question. Is $q = 8$ the right working point, and in the regimes where the scheme itself is inaccurate, does the label-free network reproduce the scheme’s own error?

Design. Vary q over $\{2, 4, 8, 16\}$ and nothing else: same population, same fiducial cut, same 4×50 network, same protocol, same split seed. Both losses, three seeds each, 24 runs. For each q the exact scheme is also solved by root-finder on the very same 2018 test states, so the ceiling is like for like.

The first two rows are the most informative result in the campaign. At $q = 2$ the label-free network lands at $5202\mu\text{m}$ against a ceiling of $5198\mu\text{m}$, within 0.1 percent. At $q = 4$ it lands at $5371\mu\text{m}$ against $5401\mu\text{m}$, within 0.6 percent. The network is not failing. It is solving the

Table 4: Stage-count sweep on the frozen cross-magnet leg. Test-split median endpoint error in micrometres, median over three seeds with the seed range in brackets. The ceiling is the same q -stage scheme solved directly by a root-finder on these same 2018 test states. The straight line is $520,444\ \mu\text{m}$ at every q . All 24 runs converged. Source: `Stage_count_sweep/results/error_vs_stages.csv`.

stages q	physics loss	data twin	exact scheme	exact scheme, p95
2	5202 [5199–5218]	174 [168–223]	5198	37,728
4	5371 [5352–5394]	181 [157–181]	5401	54,057
8	181 [165–240]	176 [162–204]	23	1610
16	187 [186–280]	170 [147–176]	20	325

equations it was given, and those equations are 5 mm wrong on this leg. This is the cleanest available demonstration that the physics loss of Eq. (13) does what it claims: it drives the network onto the solution of the discrete scheme, whatever that solution happens to be. The supervised twin at the same q sits at 174 and 181 μm , thirty times better, because its labels are the true trajectory and it never sees the scheme at all.

Between $q = 4$ and $q = 8$ the ceiling falls from 5401 μm to 23 μm and crosses below the network’s own floor. From there the limiting factor is the network and the optimiser rather than the discretisation, and $q = 16$ agrees with $q = 8$ within the seed spread, 187 against 181 μm , while costing 68 outputs per sample instead of 36. That fixes $q = 8$ as the working point.

Two things were not expected. First, $q = 4$ is no better than $q = 2$ and is slightly worse at the ceiling, 5401 against 5198 μm . Formal order $2q$ is an asymptotic statement about small steps, and a single 5178 mm step through a magnet is nowhere near that regime; the extra stages buy nothing until there are enough of them to resolve the field variation along the leg, and between four and eight there suddenly are. The root-finder converged to a residual below 10^{-8} on 99.9 percent of legs at every q , so these are genuine solutions of the discrete equations. The equations are simply far from the differential equation. Second, $q = 16$ is not the hard case: it trains faster than $q = 8$, 113 to 115 restarts against 154. The hardest optimisation in the grid is $q = 4$.

The 29 μm correction. A ceiling must always be quoted on the population being scored. An earlier measurement of the exact scheme, made on 32 legs drawn stratified in momentum and each on its own start plane, gave 29 μm at $q = 8$, and that number was quoted in July. Stratifying in momentum deliberately over-weights the soft, hard-bending tracks, so the stratified figure describes a harder population than any test split used here. On the 2018 frozen-leg test states with their natural momentum mix the ceiling is 23 μm , and the pre-stated ceilings of about 14 mm at $q = 2$ and 4.3 mm at $q = 4$ become 5198 and 5401 μm . Both curves are carried in Figure 3 so that the difference is visible rather than argued about. The qualitative claim survives intact: the physics network tracks the ceiling at low q and leaves it at $q = 8$.

5.3 Network size, seeds, and what sets the floor

Question. The frozen-leg network stops at about 180 to 220 μm while the scheme it is solving reaches 23 μm . Is the gap a limit of the network’s representational capacity, or of the optimiser’s ability to find the good weights? And which network should the later experiments use?

Design. The July experiment with two things varied: width in $\{32, 50, 100, 200\}$ crossed with depth in $\{2, 4, 6\}$, and seeds 0 to 9. That is 120 label-free runs, plus 10 supervised runs at the winning architecture, so 130 runs.

Four things follow from Table 5.

The floor is set by the optimiser, not by capacity. Growing the network from 2436 to 129,036 parameters, a factor 53, moves the median from 1059 μm to 115 μm , a factor 9.2, and the returns collapse along the way: the last width doubling, 4×100 to 4×200 , buys 14 percent for

Stage-count sweep on the frozen magnet crossing (median over 3 seeds, bars = seed spread)

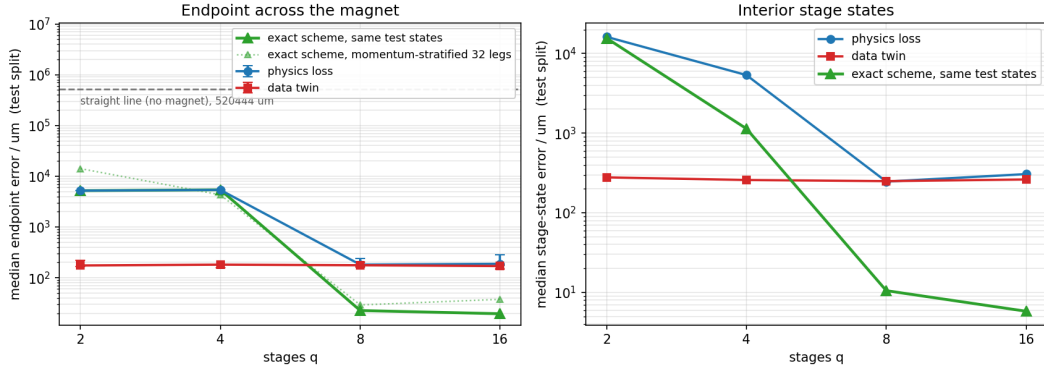


Figure 3: Median test-split endpoint error against the number of Gauss-Legendre stages q , on a logarithmic vertical axis, for the frozen cross-magnet leg. The left panel is the endpoint of the step; the right panel is the same measure taken over the eight interior stage states. Blue is the label-free physics loss, red the supervised twin, solid green the exact scheme solved by root-finder on the very same test states, and dotted green the same scheme measured on the momentum-stratified sample of 32 legs, which over-weights the hardest tracks and gives a different and non-comparable number. Bars are the spread over three seeds. The dashed horizontal line at $520,444 \mu\text{m}$ is the straight-line baseline. The blue and green curves lie on top of one another at $q = 2$ and $q = 4$, which is the physics loss reproducing the scheme’s own error; they separate at $q = 8$, where the scheme’s error drops below the network’s floor and the network becomes the limiting factor. The red curve is flat at about $175 \mu\text{m}$ at every q , because the supervised twin never sees the scheme.

Table 5: Architecture grid on the frozen cross-magnet leg at $q = 8$, ordered by parameter count. Test-split median endpoint error in micrometres over the converged seeds, with the best and worst converged seed. Unconverged runs are never pooled into a median. The last row is the supervised twin at the winning architecture. Source: `Network_size_and_seed_study/results/by_architecture.csv`.

loss	depth $\times$ width	parameters	converged	median	min	max	median final loss
physics	2×32	2,436	5/10	1059	853	1600	1.83×10^{-4}
physics	2×50	4,686	9/10	470	340	593	4.54×10^{-5}
physics	4×32	4,548	3/10	411	387	450	5.37×10^{-5}
physics	6×32	6,660	4/10	427	391	2725	6.51×10^{-5}
physics	4×50	9,786	10/10	222	165	244	1.06×10^{-5}
physics	2×100	14,336	10/10	223	201	299	1.08×10^{-5}
physics	6×50	14,886	6/10	326	248	525	2.74×10^{-5}
physics	4×100	34,536	10/10	133	128	156	3.04×10^{-6}
physics	2×200	48,636	10/10	153	128	241	4.19×10^{-6}
physics	6×100	54,736	10/10	191	152	242	9.33×10^{-6}
physics	4×200	129,036	10/10	115	100	134	2.06×10^{-6}
physics	6×200	209,436	9/10	159	150	208	6.05×10^{-6}
data twin	4×200	129,036	10/10	128	105	159	1.34×10^{-6}

3.7 times the parameters and 1.8 times the wall time. The decisive evidence is the correlation. Over all 96 converged label-free runs the Spearman rank correlation between the final training loss and the test endpoint error is $+0.965$. Every architecture and every seed lies on one band, three decades of loss mapping monotonically onto one and a half decades of error. The endpoint error is a readout of how far the optimiser got, not of what the network could represent. A wider network helps only because L-BFGS gets further down on it: the 4×50 networks stall at a median final loss of 1.06×10^{-5} , the 4×200 networks at 2.06×10^{-6} . It follows that anything improving the conditioning of the optimisation problem should be expected to pay more than

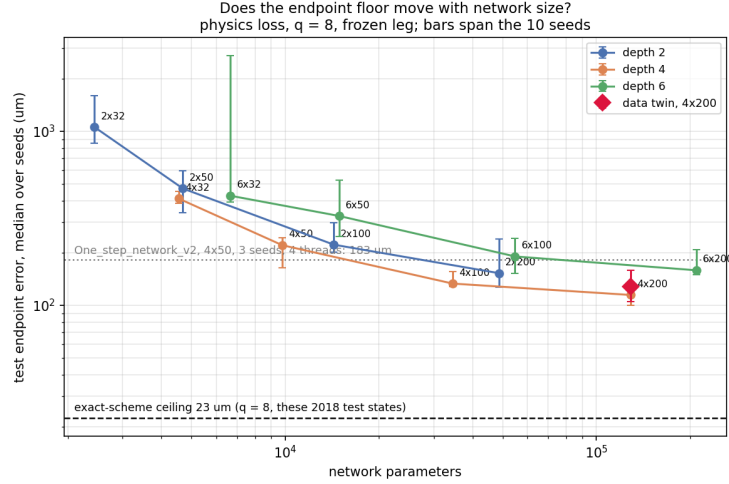


Figure 4: Median test endpoint error against the number of network parameters, both axes logarithmic, for the label-free loss on the frozen magnet crossing at $q = 8$. Each point is one architecture, labelled depth $\times$ width; the three coloured curves are depths 2, 4 and 6, and the bars span the ten seeds. The red diamond is the supervised twin at 4×200 . The dotted horizontal line marks the July three-seed baseline at $183 \mu\text{m}$ and the dashed line at the bottom marks the exact-scheme ceiling of $23 \mu\text{m}$ on these same 2018 test states. Depth 4 lies below depths 2 and 6 at every width, and the depth-4 curve flattens above about thirty thousand parameters, so that the last factor 3.7 in size buys 14 percent in error. The gap between the flattening curve and the ceiling is the factor 5 that this study attributes to the optimiser rather than to capacity.

more parameters do, which is the prediction Section 5.6 tests.⁵

Depth 4 is the optimum, and it is interior. Depth 6 is never better than depth 4 and is 39 to 47 percent worse at widths 50, 100 and 200; it is also much harder to train, with 4, 6, 10 and 9 seeds of ten converging at the four widths against 3, 10, 10 and 10 at depth 4. Depth 2 is worse than depth 4 at every width, by 33 percent at width 200 rising to 158 percent at width 32.

At the largest size the label-free loss beats its own supervised twin. At 4×200 the physics loss gives $115 \mu\text{m}$ [100–134] against the twin’s $128 \mu\text{m}$ [105–159], so the twin’s median is 12 percent higher. The twin also has the wider seed spread, a factor 1.51 between its best and worst seed against 1.33, while reaching its stall in about half the restarts. The factor-two gap that the July experiments chased is gone. With the fiducial requirement in place and a network large enough for the optimiser to work with, a loss that never sees a label matches and slightly beats supervision on labels computed with the same field.

Seeds must be pooled, and validation selection is safe for the one-step task. Over architectures with at least three converged seeds, the median ratio between the luckiest and unluckiest seed is 1.54. It shrinks with size, to 1.22 at 4×100 and 1.33 at 4×200 . Choosing a seed on validation endpoint error is reliable here: the rank correlation between validation and test endpoint error over the 96 converged runs is +0.995, the validation-best seed is also the test-best seed in 9 of 12 architectures, and where it is not, the penalty is at most 2.3 percent. Section 5.5 shows that this stops being true the moment the network is chained.

The recommendation this experiment produced, and which the general-leg experiment then only partly followed because it was already running, was 4×100 as the smallest adequate network, with 4×200 as the check that a result is not size-limited. 4×50 sits on the steep part of the curve, where a result reports the network size as much as the physics.

⁵This is the same number, to the second decimal place, that the van der Pol study measured on a two-dimensional autonomous oscillator with an analytic right-hand side. I did not expect two systems with that little in common to agree that closely, and I have no explanation for it beyond the shared optimiser.

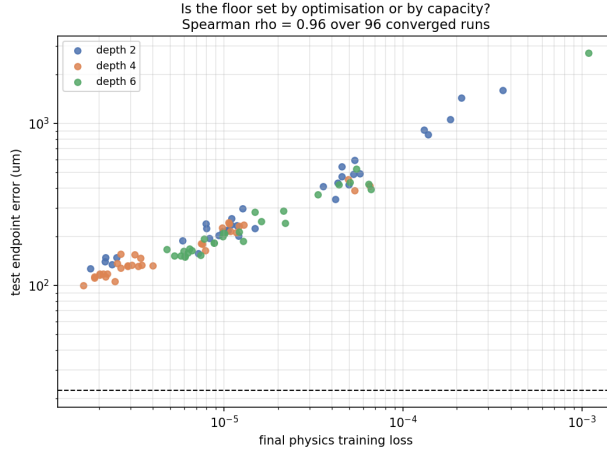


Figure 5: The evidence behind the optimisation-floor claim: final training loss against test-split endpoint error, both axes logarithmic, one point per converged label-free run, coloured by architecture. All 96 converged runs lie on a single band, three decades of loss mapping monotonically onto one and a half decades of error, with a Spearman rank correlation of $+0.965$. If the floor were a capacity limit the small architectures would sit above the band at their own loss values, forming separate branches. They do not.

Table 6: The reversed magnet polarity against the original, on the frozen cross-magnet leg, 4×50 , $q = 8$. Test-split median endpoint error in micrometres, median over seeds with the seed range in brackets. No labelled sample has ever been produced for the magnet-up polarity in this project, which is nonetheless the sample’s own polarity (Section 4.1). Source: `Magnet_up_field/results/summary.csv` and `Magnet_up_field/results/ceiling_summary.json`.

quantity	magnet up	magnet down
network, label-free physics loss	229 [171–296], 10 seeds	222 [165–244], 10 seeds
network, supervised data twin	204 [187–230], 3 seeds	192 [162–208], 3 seeds
exact-scheme ceiling on the test states	22.1	22.5
straight line	525,184	520,444
network divided by its own ceiling	$10.4 \times [7.7\text{--}13.4]$	$9.8 \times [7.2\text{--}10.8]$

5.4 Training where no labels exist: the magnet-up polarity

Question. The claim of the technique is that no labels are needed. The strongest available test of that claim is to train on a configuration for which no labelled sample has ever been produced. LHCb periodically reverses the magnet’s polarity, so there is a second field map, and no training labels have ever been made for it in this project.⁶

Design. The identical 4×50 network, the identical protocol, the identical training population, with the single change that the loss is built on the reversed map. Label-free seeds 0 to 9 plus three supervised control runs, so 13 runs, all converged. The reversed-polarity references exist only to score; the label-free runs never see them.

Three checks were run before submitting. The two maps are exactly one another’s negative: on 200,000 random in-map points the maximum of $|\mathbf{B}_{\text{up}} + \mathbf{B}_{\text{down}}|$ is 0.0 in all three components, so the field magnitude profile is bit-for-bit identical and only the sign of the bending differs. Both polarities are built from the same event rows, since the selection uses no field at all. And a probe run established that the polarity flag really reaches the loss and is not silently ignored: one optimiser restart from the same seed on the same states gives 2.62×10^{-3} with the loss built on the reversed map against 8.03×10^{-3} with it built on the original.

⁶Magnet up is in fact the polarity the sample was simulated with, established after this experiment was run; see the correction box in Section 4.1. The experiment is therefore the only one in this paper scored against the physically right field. Its numbers are unchanged; “no labels have been made for it” remains true, and “a polarity the sample is not” would not be.

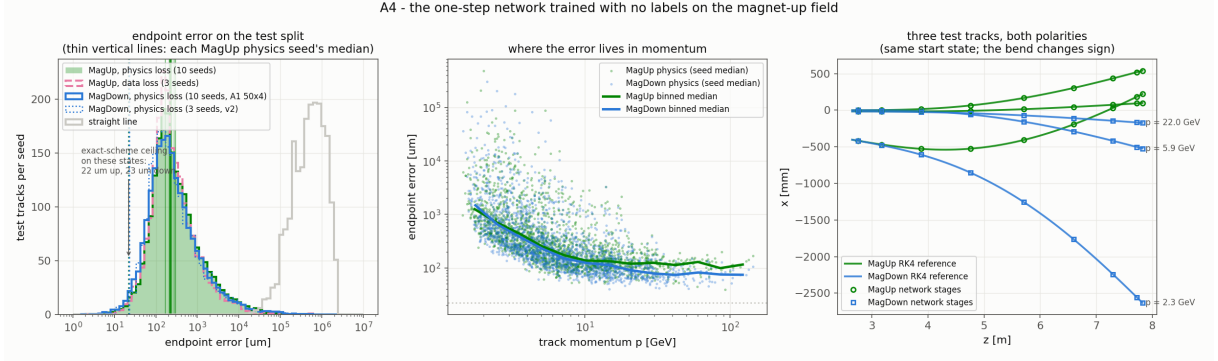


Figure 6: Three panels summarising the reversed-polarity experiment. Left: the distribution of test-split endpoint errors on a logarithmic horizontal axis, with the ten reversed-polarity label-free seeds in filled green, the three reversed-polarity supervised seeds in dashed pink, the ten original-polarity label-free seeds in solid blue, the three July original-polarity seeds in dotted blue, and the straight-line baseline in grey near one million micrometres; the two polarities lie on top of one another and the two exact-scheme ceilings, 22 and 23 μm , are marked at the left. Middle: endpoint error against track momentum with binned medians overlaid; the two polarities trace the same curve, falling from about one millimetre at 2 GeV to about 100 μm above 20 GeV. Right: three test tracks propagated from the same start states under both polarities, with the network’s eight stage states as markers on the reference curves; the trajectories curve in opposite directions, which is the check that the network learned the sign of the bending and not merely the geometry.

The label-free network reaches the same floor on the reversed field as on the original and sits the same distance above the reversed field’s own ceiling. The two ten-seed ranges overlap almost completely, and the four percent difference in medians is far inside a seed spread that is itself a factor 1.7 wide on the reversed polarity and 1.5 on the original. It also matches its own supervised twin on both, 229 against 204 μm and 222 against 192 μm . Both are three orders of magnitude below the straight line, so the network has learned the bending and not merely the geometry, and it has learned it with the correct sign.

One genuine asymmetry was found, and it is in the tracks rather than in the map. The fiducial requirement removes 21, 20 and 15 states from the three splits on magnet down and *none* on magnet up: the sample’s charge and angular distribution is not symmetric in x , and magnet down is the one that bends these particular soft tracks out of the map while magnet up bends them back in. The same asymmetry appears in the exact scheme’s worst case, 38 mm reversed against 258 mm original. A control that re-scores the reversed networks on only the 2018 tracks the original polarity also keeps moves the median from 229.4 to 226.6 μm , about one percent, so the medians are unaffected. Statements about tails, however, must say which polarity they refer to.

5.5 One network for every kind of step, with absolute outputs

Question. Everything so far trains a network on one frozen leg: every sample starts on the same plane and ends on the same plane, so the network cannot be used anywhere else. Two things have to be true before this is a usable extrapolator. One network must serve steps of arbitrary length starting anywhere, and it must survive being applied repeatedly to its own output.

Design. The leg becomes an input. Each sample carries its own start plane z_0 and its own step length Δz , normalised and handed to the network as two extra inputs, and the stage planes become per-sample quantities $z_0 + c_j \Delta z$ rather than constants. One network is asked to serve leg types A, B and C at once. The training size was fixed by a rule stated before any result was looked at, namely the largest of 2000, 4000, 8000 and 16000 for which one optimiser restart runs in under 90 seconds, giving $N = 8000$, or 7935 train, 3969 validation and 3961 test states after

Table 7: One network for legs A, B and C with absolute outputs. Test-split median endpoint error in micrometres, median over converged seeds, with the seed range in brackets where quoted. The ceiling is the exact $q = 8$ scheme solved on this experiment’s own test states, which is why it differs from the stratified numbers of Section 5.2. The 4×200 column excludes one unconverged physics seed. Source: `General_leg_network/results/by_leg.csv`.

leg	physics loss			data twin			straight	ceiling
	4×50	4×100	4×200	4×50	4×100	4×200		
A vertex fetch, 341 mm	849	404	229	653	385	227	10.9	0.0013
B cross-magnet, 5175 mm	6742	2600	1740	2395	933	777	444,075	46
C plane-to-plane, 70 mm	535	300	213	486	235	182	6.6	6×10^{-7}

the fiducial cut. The leg mix is the training set’s own and is not stratified: 1317 of type A, 1060 of type B and 5558 of type C in the train split. Architectures 4×50 and 4×100 , label-free seeds 0 to 9, supervised seeds 0 to 2, so 26 runs, all converged. A second wave at 4×200 followed overnight, 13 runs of which 12 converged.

One label-free network does serve all three leg types, and it does so nowhere near any of their ceilings. On the cross-magnet leg the best label-free network at 4×100 sits at $2600 \mu\text{m}$ against a $46 \mu\text{m}$ ceiling, a factor 56. Compared with the frozen-leg network of Section 5.1 on the same kind of leg, $183 \mu\text{m}$, making the leg an input costs a **factor 14**, and it costs a factor 11.7 against the ten-seed frozen-leg figure of $222 \mu\text{m}$ even though the general network is twice as wide. On the two short leg types the network is worse than doing nothing: ignoring the magnet entirely gives $10.9 \mu\text{m}$ on leg A and $6.6 \mu\text{m}$ on leg C, while the network gives $404 \mu\text{m}$ and $300 \mu\text{m}$.

Width keeps paying on the one-step score, with clear diminishing returns. Leg B goes $6742 \rightarrow 2600 \rightarrow 1740 \mu\text{m}$ for widths $50 \rightarrow 100 \rightarrow 200$, factors of 2.6 then 1.5 for each doubling, and is still 38 times above the $46 \mu\text{m}$ ceiling. On the short legs the widest network is still 32 times (leg C) and 21 times (leg A) worse than ignoring the magnet entirely, which no amount of width can fix.

The supervised twin is uniformly ahead here, by a factor 2.8 on leg B at 4×100 and by 5 to 25 percent on legs A and C, where on the frozen leg the two were indistinguishable. The gap between the two losses opens exactly where the leg geometry varies. Where the error sits inside the step is also informative: on legs A and C it is flat across the eight Gauss nodes, which is the signature of a scale error rather than an accumulation, while on leg B it grows monotonically from the first node to the endpoint, roughly doubling, which is the bend being under-resolved.

5.5.1 Chaining: the network walked along real particle paths

For each test particle I collect the planes named by its stored forward legs, sort them, and call consecutive planes a leg. The chain starts from the state at the first plane, and the truth is the double-precision Runge–Kutta path from that same start state through the same planes, so the network and the reference begin in the same place and only the propagation is compared. Kept: 4563 test particles with four or more legs, 23,500 legs in total, median leg length 70 mm and 99th percentile 5.2 m. All trained networks were walked along all of them.

The obvious rule, that row i is followed by row j when the end plane of one is the start plane of the other, was tried first and is not usable: only 15 of the 7636 test particles have four rows that butt together exactly, because the training set stores at most three plane-to-plane legs per particle and they need not be adjacent. Using the sorted planes as waypoints keeps every stored plane and fills the gaps with one more leg of the same kind, which is what a real extrapolator has to do anyway.

Nothing breaks: no chain diverges, and the 95th percentile grows at the same rate as the median. But nothing saturates either. The step-to-step growth ratio is 3.2, 1.8, 1.6, 2.1, 1.8, 1.1 at 4×100 with the label-free loss, and essentially the same at every architecture and both losses,

A3a one network for legs A, B and C: test-split endpoint error by leg type and momentum

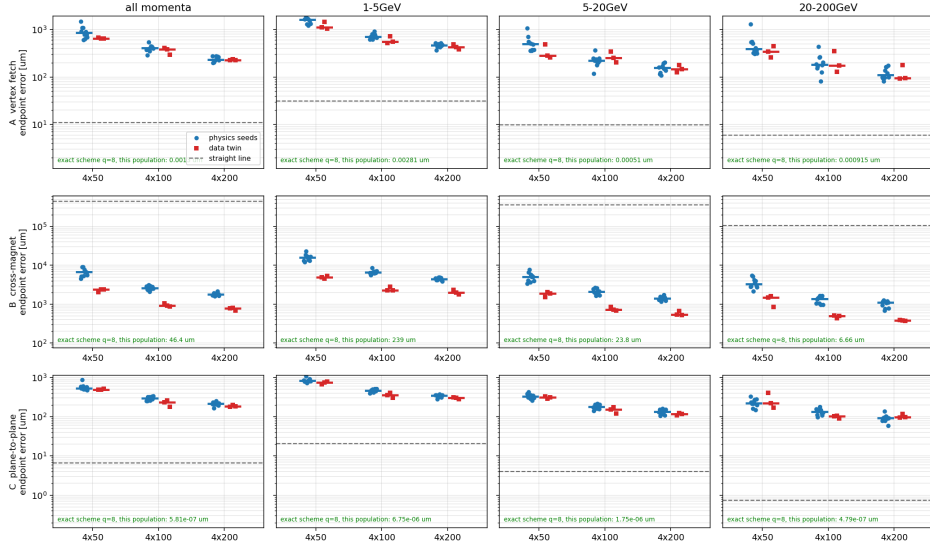


Figure 7: Test-split endpoint error for the single general network with absolute outputs, by leg type (rows: A vertex fetch, B cross-magnet, C plane-to-plane) and momentum band (columns: all momenta, then 1 to 5, 5 to 20 and 20 to 200 GeV), on a logarithmic vertical axis. Blue circles are the ten label-free seeds and red squares the three supervised seeds, with horizontal bars at the median; the dashed grey line in each panel is the straight-line baseline and the green annotation gives the exact-scheme ceiling on that same population. The middle row shows the network beating the straight line by two orders of magnitude on the cross-magnet leg while sitting a factor 56 above the ceiling; the top and bottom rows show the opposite situation, where the straight line lies below every network point.

Table 8: The general network with absolute outputs, chained along real particle paths. Median endpoint error against the reference path, test split, in micrometres, by the number of legs walked. All 4563 particles contribute up to leg 4; beyond that only the longer chains do, so the first four rows are the like-for-like comparison. Source: `Chained_legs/results/chain_summary.csv`.

legs walked	physics loss			data twin			particles
	4×50	4×100	4×200	4×50	4×100	4×200	
1	440	246	174	388	217	165	4563
2	1423	783	540	1275	652	510	4563
3	2562	1443	1029	2225	1051	894	4563
4	3919	2293	1738	3634	1484	1490	4563
5	8163	4831	4133	6991	2869	2972	2914
6	14,479	8904	8659	10,087	4820	5695	1481
7	15,999	10,248	10,450	10,739	5283	6476	853

so the error roughly doubles per leg. That is not the square-root growth of independent random errors; it is systematic, each leg starting from a state that is already displaced and adding its own bias on top. Over a full particle path the network is 5 to 10 mm out. Width shifts the curve down and does not change its slope: at seven legs the 4×200 physics chain ends at $10,450 \mu\text{m}$ against $10,248 \mu\text{m}$ for 4×100 , which is the sense in which chained error is not bounded.

Chaining also changes how a seed should be chosen. The cheap one-step validation error, which Section 5.3 showed to be an excellent selector for the one-step task, carries no information about chained performance: at 4×100 its rank correlation with the test chain error is 0.08, and the best one-step seed is only sixth best when chained. The validation *chain* error transfers to test almost perfectly, rank correlation 0.92 and 0.95 at the two architectures and 0.94 at 4×200 . A seed must be selected on the quantity that will actually be used.

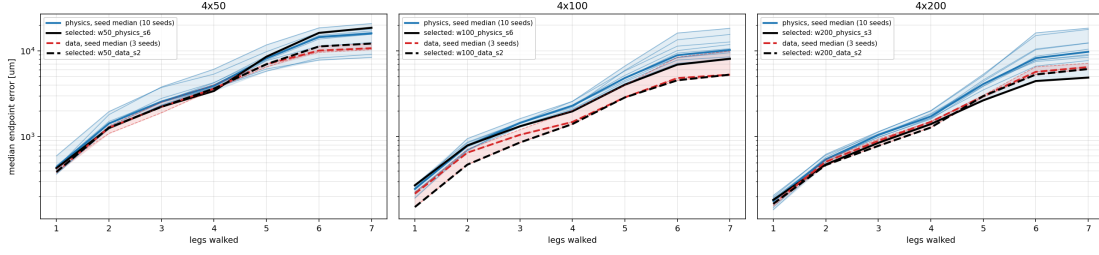


Figure 8: Median endpoint error against the number of legs walked, logarithmic vertical axis, for the general network with absolute outputs. Each curve is one architecture and loss; bands are the seed spread and the dashed curves are the 95th percentiles. The number of contributing particles falls after leg four, marked on the axis, because only the longer chains reach those lengths. The curves are straight on this axis over the first four legs, which is geometric growth: the error roughly doubles per leg rather than growing like the square root of the number of legs, the signature of a systematic bias rather than independent random errors.

Table 9: Leg D, the giant backward step to the collision vertex, over 2726 test particles, in micrometres, median over converged seeds. No leg of this geometry was in any training set. Source: `Chained_legs/results/leg_d_reproduction.csv` and `leg_d_ceiling_same_population.csv`.

	physics loss			data twin		
	4×50	4×100	4×200	4×50	4×100	4×200
two composite steps	47,353	59,416	35,034	34,228	33,416	30,351
the same leg in one giant step	17,947	8778	4563	6213	3158	2182
exact $q = 8$ scheme on these legs: 752; $q = 16$: 216; straight line: 948,000						

5.5.2 Leg D: can a hard step be replaced by two easier ones?

Leg D, the downstream leg from the first post-magnet plane all the way back to the collision vertex, is a single backward step with a median length of -8.6 m. No leg D was in the training set, so this is an out-of-training test by construction. The exact eight-stage scheme itself is only good to $752 \mu\text{m}$ on it, and $216 \mu\text{m}$ at $q = 16$; the straight line is 948 mm.

Composite stepping is four to ten times *worse* than the single giant step, not better, and both are more than an order of magnitude above the scheme’s ceiling. The reason is visible in the first hop, which is longer than any leg in training and is already 1.5 to 10 mm out. This is not a sign problem: the network has seen backward steps, since all 1317 A legs and 490 of the 1060 B legs are backward. It is that no leg-D geometry was in training at all, and the network does not extrapolate to it. It does still beat the straight line by a factor 100 to 300, which is the only thing that can be said for it here.

5.5.3 The hypothesis this experiment produced

I stated the following as a hypothesis before testing it. The general-leg network is limited by its output parameterisation: it predicts the **absolute** state at each stage and at the endpoint, scaled by a single population-wide constant per component. Four pieces of evidence pointed that way.

1. The output scale on the general-leg dataset is 602 mm in x and 341 mm in y , set by the spread of the whole population, which is dominated by the 5 m cross-magnet legs. The physical correction the network must produce on a 70 mm plane-to-plane leg is the straight-line error on that leg, about $7 \mu\text{m}$. Resolving a $7 \mu\text{m}$ correction on an output whose scale is 602 mm is asking for a relative precision of one part in 10^8 from a network whose loss

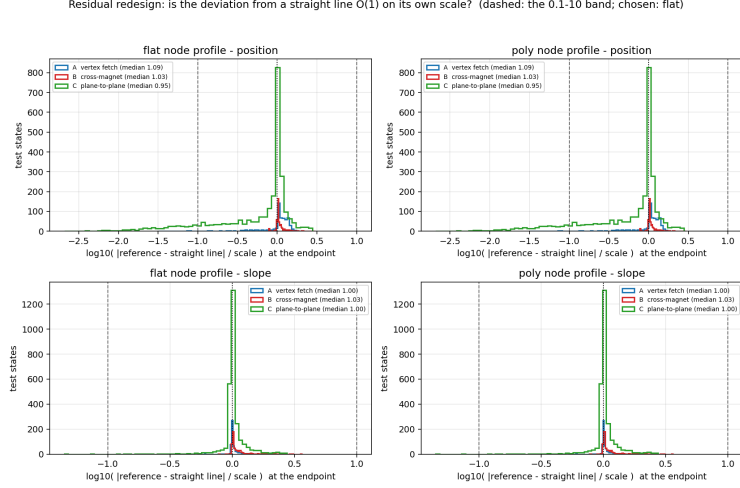


Figure 9: The check that had to pass before any farm time was spent: the distribution, by leg type, of the ratio the network will actually have to emit, that is the reference deviation from a straight line divided by the label-free scale of Eq. (15). The two node profiles considered are shown side by side, flat and polynomial. The rule, fixed before looking, was to take flat unless its endpoint medians fell outside 0.1 to 10 on some leg while the polynomial profile’s did not. The medians are 1.09, 1.03 and 0.95 on legs A, B and C for both profiles, so the rule selects flat. The same label-free expression lands within ten percent of the truth on a 341 mm backward vertex fetch, a 5.2 m magnet crossing and a 70 mm sensor hop, which is the $O(1)$ requirement the redesign was built around.

is normalised at the population scale. On the frozen-leg dataset, where the population is one geometry, the corresponding output scale is 396 mm and the correction is 520 mm, a completely different regime.

2. The error is roughly a fixed fraction of the largest step the network was trained on, not a fixed fraction of each step: 404, 2600 and 300 μm on legs whose true corrections span five orders of magnitude, from 6.6 μm to 444 mm.
3. On the short legs, where the hypothesis says the error should be a scale error rather than an accumulation, the error is flat across the eight Gauss nodes. On the long leg, where the bend is genuinely being under-resolved, it grows monotonically to the endpoint.
4. The frozen-leg network, which has exactly the same architecture, loss, optimiser and data source but only one geometry and therefore a matched scale, is a factor 14 better on that geometry.

The alternative was capacity, and the second wave at 4×200 was run to separate them. Section 5.3 predicted that width would buy about 14 percent over 4×100 on the frozen leg; on the general-leg dataset it bought a factor 1.5 on leg B, leaving the factor 14 largely intact and the short legs still an order of magnitude worse than a straight line. That is consistent with the parameterisation hypothesis and not with the capacity one.

5.6 The straight-line-residual redesign

Design. Exactly one thing changes: what the network is asked to output, per Eq. (14), with the per-leg scale of Eq. (15) built from the bending integral of Eq. (16). The dataset is byte for byte the same population plus three scale arrays per split. The physics loss, the optimiser protocol, the number of stages, the splits, the momentum bands, the scorer and the ceilings are all unchanged. Architectures 4×50 and 4×100 , label-free seeds 0 to 9, supervised seeds 0 to 2: 26 runs, all 26 converged.

Table 10: The residual redesign against the absolute-output design, by leg type. Test-split median end-point error in micrometres, median over seeds, with the seed range in brackets. The residual 4×50 ranges, omitted for width, are 3–6, 945–1627 and 1–1. Every column here was re-scored in one pass on the same test states by `aggregate_residual.py`. Source: `General_leg_network/results/by_leg_residual.csv`.

leg	absolute, physics		residual, physics		resid.		
	4×100	4×200	4×100	4×50	twin	straight	ceiling
A vertex fetch	404 [286–538]	229 [201–277]	4.4 [2–6]	4.1	0.06	10.9	0.0013
B cross-magnet	2600 [2106–3152]	1740 [1632–2154]	1026 [878–1209]	1136	512	444,075	46.4
C plane-to-plane	300 [252–335]	213 [165–235]	1.6 [1–2]	1.2	0.01	6.6	5.8×10^{-7}

Table 11: Whole test split, all leg types pooled, median over seeds, in micrometres. The straight line on this split is $16.1 \mu\text{m}$. The absolute-output design never beat it at any width; the residual design beats it by a factor 7.6 with the physics loss and 410 with the twin, and a 4×50 residual network beats a 4×200 absolute-output network by a factor 129. Source: `General_leg_network/results/residual_summary.csv`.

output design	physics loss			data twin		
	4 × 50	4 × 100	4 × 200	4 × 50	4 × 100	4 × 200
absolute states	721	397	273	637	321	237
straight-line residual	2.12	2.64	not run	0.06	0.04	not run

The initialisation check. Before the runs, `train_residual.py --init-check` verified the algebra at 4×100 , seed 0, on the 7935 training states: the straight-line term agrees with an independent construction to 4.5×10^{-13} mm; zeroing the last layer makes the output minus the straight line exactly 0.0 mm; at the shared trainer’s own initialisation the raw network output has median 0.057 and maximum 0.34 in units of the residual scale, which is $0.51 \mu\text{m}$ at the median endpoint; and both losses and their maximum gradients are finite. The last layer is *not* zeroed for the real runs, because a zero last layer makes every earlier layer’s gradient exactly zero on the first step; zeroing is used only as the algebra check. The consequence is worth stating: an *untrained* residual network already scores $15.5 \mu\text{m}$ on the whole test split, against $16.1 \mu\text{m}$ for a straight line, where the converged absolute-output networks scored about $300 \mu\text{m}$.

The optimisation got easier, not just the answer better. The residual runs stall sooner, 40 to 64 restarts against 84 to 121 at the same width, at a physics loss an order of magnitude lower, 9×10^{-7} against 1.8×10^{-5} at 4×100 seed 0. Only three of the 26 runs needed the resubmission treatment, against 20 of the first 26 in the absolute-output wave.

Read across Table 10.

- The two short legs are transformed.** Leg A goes $404 \rightarrow 4.4 \mu\text{m}$, a factor 92; leg C goes $300 \rightarrow 1.6 \mu\text{m}$, a factor 190. Both now sit *below* the straight line, by a factor 2.5 on A and 4 on C, where the absolute-output design sat thirty to fifty times above it, which is the hypothesis’s own prediction.
- The cross-magnet leg improves by a factor 2.5,** $2600 \rightarrow 1026 \mu\text{m}$. Real, but an order of magnitude short of the same transformation. Leg B is the one leg where the absolute output scale was roughly right, so it had least to gain. The redesign fixed a conditioning problem, and what is left on leg B is not a conditioning problem.
- Width has stopped mattering** on legs A and C: 4×50 and 4×100 are within 20 percent of each other, and on both legs the narrower network is the better of the two. The absolute-output verdict read the width trend as evidence that width was the binding constraint. It was never width. It was the parameterisation.

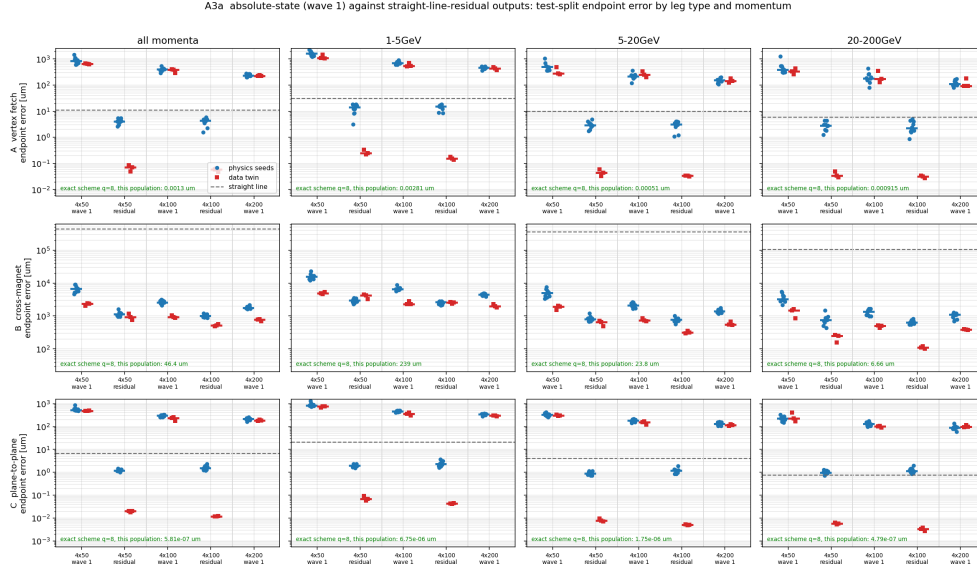


Figure 10: The same layout as Figure 7, with the residual arm added. Rows are leg types A, B and C, columns are momentum bands, and the vertical axis is logarithmic. The absolute-output seeds are the upper cloud in each panel and the residual seeds the lower one, with the straight-line baseline as a dashed grey line and the exact-scheme ceiling annotated. On the top and bottom rows the residual cloud crosses from above the straight line to below it, which is the whole point of the redesign; on the middle row it moves down by a factor 2.5 and stays well above both the ceiling and its own supervised twin. The improvement on legs A and C is uniform in momentum, a factor 46 to 200 in every band, so it is not an artefact of the momentum mix. The one cell where the residual network is still worse than a straight line is plane-to-plane at 20 to 200 GeV, $1.2 \mu\text{m}$ against $0.74 \mu\text{m}$: a stiff 70 mm hop is so nearly straight that there is essentially nothing to predict, and the network's own noise floor is above the thing it is correcting.

4. **The data twin is extraordinary on the short legs**, $0.06 \mu\text{m}$ on A and $0.01 \mu\text{m}$ on C, four orders of magnitude better than the absolute-output twin on each, and still twice as good as the physics loss on leg B, 512 against $1026 \mu\text{m}$, where it is a factor 11 above the ceiling. The gap between the two losses has widened in relative terms on the easy legs and narrowed on the hard one.
5. **Distance to the ceiling, which is the honest scorecard.** Against the exact $q = 8$ scheme on the same states, the residual 4×100 physics network is 3400 times above the ceiling on leg A, 22 times on leg B and 2.7×10^6 times on leg C. The absolute-output factors were 310,000, 56 and 5×10^8 . So the gap closed by two orders of magnitude on A and C and by a factor 2.5 on B, but the network is nowhere near the scheme it is solving on any leg, and on the short legs the scheme is so nearly exact that no network is going to reach it.

5.6.1 Where the error lives inside the step

Table 12 is the clearest single piece of evidence that the diagnosis of Section 5.5.3 was right.

5.6.2 What did not improve: the tails

A small population of states got worse while the bulk got thousands of times better. Those are the soft tracks that bend hardest. The residual scale of Eq. (15) is a first-order estimate, and where the true deviation is many times that estimate the network is back to writing down a large number. Fixing the median did not fix the tail, and the tail is what an extrapolator with a bounded time budget actually has to survive.

Table 12: Median position error at each of the eight Gauss nodes and at the endpoint, test split, 4×100 physics loss, median over seeds, in micrometres. The absolute-output error on legs A and C is flat across the step, which is the signature of a scale error the network cannot resolve. The residual error instead grows monotonically from the first node to the endpoint, by a factor 50 to 70 on A and C: the flat offset is gone and what is left is the honest accumulation of an under-resolved bend. Leg B kept its monotonic growth and simply scaled down by 2.5. Source: `General_leg_network/results/stage_errors_residual.csv`.

	node 1	2	3	4	5	6	7	8	endpoint
absolute, leg A	365	340	353	320	312	408	349	381	404
residual, leg A	0.06	0.10	0.43	1.01	1.74	2.81	3.64	4.34	4.42
absolute, leg C	289	240	249	270	260	271	268	278	300
residual, leg C	0.03	0.05	0.29	0.65	0.88	1.12	1.34	1.52	1.57
absolute, leg B	987	905	1054	1436	1907	2326	2536	2519	2600
residual, leg B	108	143	277	456	616	812	924	988	1026

A3a error along the step, both arms, all widths (test split)

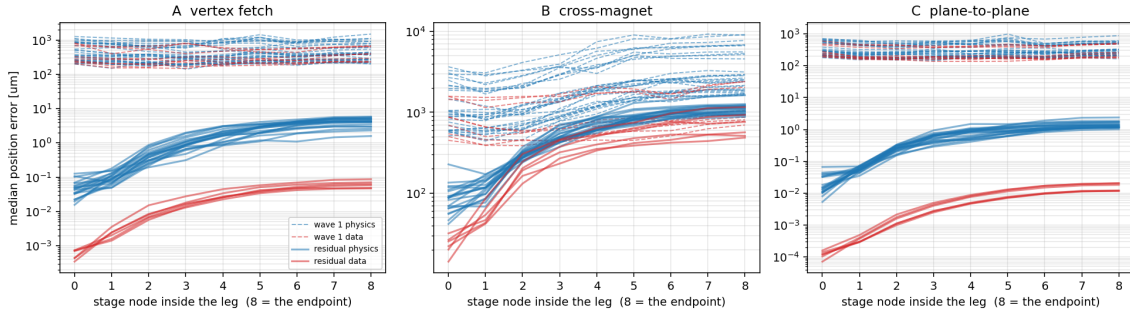


Figure 11: The error along the step, both output designs, one panel per leg type. The horizontal axis is the eight Gauss–Legendre nodes and then the endpoint; the vertical axis is the median position error in micrometres, on a logarithmic scale. The absolute-output curves on legs A and C are flat, the residual curves rise steeply from a very small first node. A flat profile means the error is present before any physics has happened and is therefore an offset; a rising profile means it is accumulated along the step.

5.6.3 Chaining the residual networks

Two things are true at once and they pull in opposite directions.

The chain is better everywhere. After four legs the residual network is 4.0 times better than the absolute-output one with the physics loss, 572 against 2293 μm , and 12.6 times better with the twin, 118 against 1484 μm . Over a full seven-leg path it is still 2.3 and 4.2 times better. The 95th percentile improves by the same factor at every step, 13.1 mm against 38.6 mm at leg four, so this is the whole distribution moving and not the median alone.

But the compounding got worse. The step-to-step growth ratios at 4×100 with the physics loss are 48.6, 4.67, 2.04, 3.06, 2.17, 1.18, against 3.18, 1.84, 1.59, 2.11, 1.84, 1.15 for the

Table 13: The 95th percentile of the endpoint error, all momenta, 4×100 , in micrometres. Leg C’s tail improves with its median by a factor 10 to 70. Legs A and B do not, and the residual twin’s tail on leg A is twice the absolute-output twin’s. Source: `General_leg_network/results/by_leg_residual.csv`, column `endpoint_p95_um`.

leg	absolute, physics	residual, physics	absolute, twin	residual, twin
A	30,693	23,640	10,445	20,128
B	41,542	27,080	18,395	17,425
C	2862	292	2193	32

Table 14: The residual networks chained along the same real particle paths, and the absolute-output networks for comparison. Median endpoint error against the reference path, test split, in micrometres. Source: `Chained_legs/results/chain_summary_residual.csv`.

legs	absolute 4×100		residual			
	walked	physics	twin	4×100	4×100 twin	4×50 particles
1		246	217	1.2	0.01	1.1
2		783	652	60	2.7	59
3		1443	1051	281	32	261
4		2293	1484	572	118	568
5		4831	2869	1750	471	1919
6		8904	4820	3804	1052	3973
7		10,248	5284	4492	1259	4739

Table 15: Leg D, the giant backward step to the collision vertex, under the residual parameterisation, over the same 2726 test particles, in micrometres. The stored label agrees with a fresh reference pass to $0.045\mu\text{m}$. Source: `Chained_legs/results/leg.d_residual.csv`.

	absolute 4×100		residual		
	physics	twin	4×100	4×100 twin	4×50
two composite steps	59,416	33,416	4252	2374	4768
one giant step	8778	3158	3749	2213	3806
exact $q = 8$ scheme on these legs: 752; $q = 16$: 216; straight line: 948,000					

absolute-output design. The residual network starts two hundred times lower, $1.2\mu\text{m}$, which is essentially its own one-step plane-to-plane number, and then multiplies by 49 on the second leg and 4.7 on the third before settling into the same factor of two per leg. By leg four the two curves are only a factor 4 apart and by leg seven a factor 2.3.

The reason is visible in the one-step table. A chain’s early legs are short plane-to-plane hops, where the residual network is at $1.6\mu\text{m}$, and its later legs include the cross-magnet crossing, where it is at $1026\mu\text{m}$, the leg the redesign barely improved. The chain error is therefore dominated, after two or three steps, by the one leg that did not get better. The redesign bought a much better start, not a slower compounding. Nothing diverges and nothing saturates.

Seed selection survives the change. Ranking physics seeds by the median chain error after four legs on validation and reading them out on test gives rank correlations of 0.98 at 4×50 and 0.90 at 4×100 , so a seed can still be chosen on validation data without touching test. The cheap alternative improves but is still not usable: the one-step validation error now correlates with the chained test error at 0.72 at 4×100 and 0.35 at 4×50 , where in the absolute-output arm it carried no information at all at 0.08. That is consistent with the picture above, since once the one-step number is dominated by the same leg that dominates the chain the two start to agree, but 0.72 over ten seeds is not enough to select on, and the ten seeds differ by 60 percent in chained error at fixed architecture, 487 to $783\mu\text{m}$ at 4×100 .

5.6.4 Leg D under the residual parameterisation

Three readings. The composite improves by a factor 14, from 59.4 to 4.3 mm at 4×100 with the physics loss, far more than the single giant step’s factor 2.3: the composite’s first hop is a -6.4 m backward step, longer than any B leg, and it went from 4.1 to 1.5 mm, while its second hop is a vertex-fetch-like leg, exactly where the residual arm gained a factor 92 one-step. Walking the leg in two steps is still worse than taking it in one, 4252 against $3749\mu\text{m}$, but the gap has closed from a factor 6.8 to a factor 1.13, so the earlier conclusion that composite stepping actively destroys accuracy is now only marginally true. And both remain five times the exact scheme’s own ceiling of $752\mu\text{m}$ on this geometry, with the 95th percentile barely moved, 142 mm against

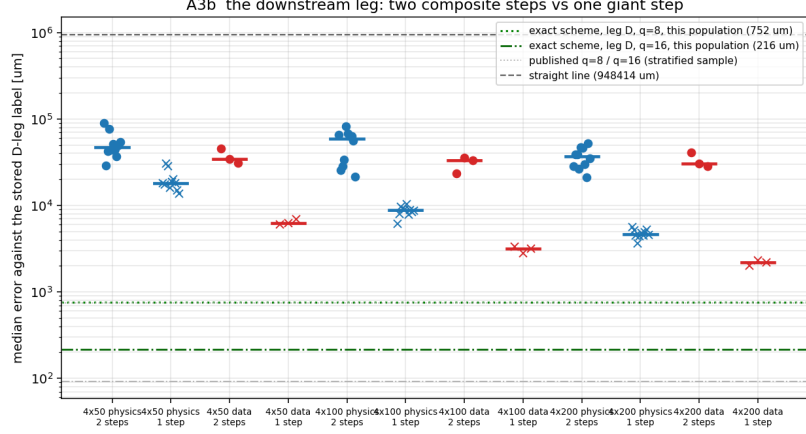


Figure 12: Leg D, the single backward step of median length -8.6 m from the first post-magnet plane to the collision vertex, a geometry that appears in no training set. Left: the distribution of endpoint errors for the composite route and for the single giant step, on a logarithmic axis, with the exact scheme’s ceiling and the straight-line baseline marked. Right: the same comparison against track momentum. The composite route, which was supposed to replace a hard step by two easier ones, is worse than the single step at every momentum, because its first hop is longer than any leg in training.

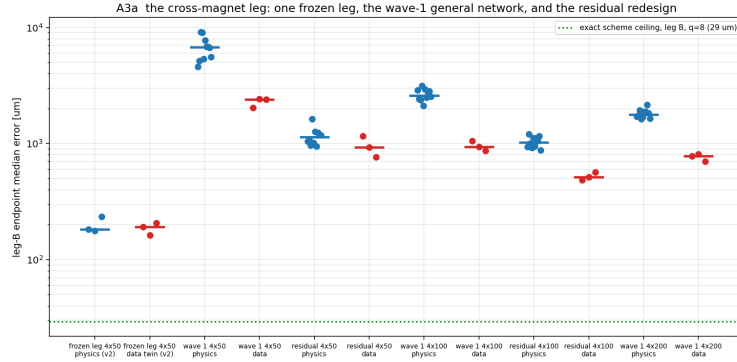


Figure 13: The cross-magnet leg across all three designs on one axis: the frozen-leg network, which trains on that geometry alone; the general network with absolute outputs; and the general network with the straight-line-residual parameterisation. The exact-scheme ceiling on each population and the straight-line baseline are drawn. Making the leg an input costs a factor 14 against the frozen-leg network, and the residual redesign recovers a factor 2.5 of it. This gap, and not the output scale, is what Section 7 proposes to attack next.

149 mm: the soft-track tail that dominates leg D is untouched.

5.7 The assembled verdict

Table 16 puts every measurement on one page. All entries are test-split median endpoint errors in micrometres, defined as in Section 3.7, with the range across seeds in brackets where more than one seed contributes. The frozen leg is the cross-magnet crossing from $z = 2648$ mm to $z = 7826$ mm with $q = 8$ stages.

The cost argument. For the same accuracy class, solving the exact eight-stage scheme with a root-finder takes a median of 44 residual evaluations per leg, each of which evaluates the field at all eight stage points, and the number is not known in advance. The trained network takes **one forward pass**, with all eight stage states emitted at once, and its cost is fixed. That ratio, and not the accuracy, is the argument for the whole exercise: a trigger can schedule a fixed cost and cannot easily schedule a loop whose trip count depends on the track.

Table 16: The whole programme on one page. Test-split median endpoint error in micrometres. Columns 1 and 2 are the frozen cross-magnet leg at two network sizes; column 3 is the same leg on the magnet-up polarity; columns 4 and 5 are the single network serving legs A, B and C, quoted per leg, under the two output parameterisations; column 6 is that network walked four legs along a real particle path. Each column’s ceiling is measured on that column’s own population.

	frozen leg		magnet up	one network, legs A / B / C, 4×100		chained
	4×50	4×200	4×50	absolute	residual	4 legs
exact scheme (ceiling)	23	23	22	0.0013 / 46 / 6×10^{-7}	0.0013 / 46 / 6×10^{-7}	n.m.
network, physics loss	222	115	229	404 / 2600 / 300	4.4 / 1026 / 1.6	572
network, data twin	192	128	204	385 / 933 / 235	0.06 / 512 / 0.01	118
straight line	520,444	520,444	525,184	10.9 / 444,075 / 6.6	10.9 / 444,075 / 6.6	n.a.
seed range, physics	[165–244]	[100–134]	[171–296]	10 seeds	10 seeds	selected seed

6 Discussion

6.1 Question 1, does it work: yes, on every test

Four measurements answer it, and they are the four the question was written around.

The label-free training is stable. Across the campaign 232 runs were trained and 208 reached a confirmed stall: all 24 of the stage sweep, all 26 of the general-leg first wave, 12 of 13 of the second wave, all 13 of the reversed-polarity experiment, all 26 of the residual redesign, and 96 of the 120 in the architecture grid. The 24 that did not confirm are all in the architecture grid, all at width 32 or depth 6, and none of them diverged: they failed the confirmation pass, in which a fresh optimiser moves a small or deep network again rather than re-stalling. No run blew up.

Where the scheme is the limiting factor, the network sits on the scheme’s own error, 5202 against $5198 \mu\text{m}$ at $q = 2$ and 5371 against $5401 \mu\text{m}$ at $q = 4$, agreements of 0.1 and 0.6 percent (Table 4). **Where the network is the limiting factor, the shortfall is identified:** optimisation and not capacity, by the loss-error rank correlation of +0.965 over 96 converged runs (Figure 5). **At 4×200 the label-free loss beats its own supervised twin,** whose median is 12 percent higher with a wider seed spread. **And on a field polarity for which no labels exist anywhere in the project** — which is, in fact, the sample’s own polarity (Section 4.1) — it reaches the same floor, 229 against $222 \mu\text{m}$, and sits the same distance above that polarity’s own ceiling (Table 6).

6.2 Question 2, can it be used: the residual design changed the answer

Under the absolute-output design the answer was mechanically yes and usefully no. The general network was a factor 14 worse than the frozen-leg network on the same kind of leg and worse than a straight line on the two short leg types; the chained error compounded to 5 to 10 mm over a full path; and composite stepping made the giant downstream leg worse rather than better.

The residual redesign of Section 5.6 changed three of those four statements. The short legs went from thirty to fifty times worse than ignoring the magnet to two to four times better, and they did so uniformly in momentum. The whole test split went from $274 \mu\text{m}$ at the best absolute-output width to $2.1 \mu\text{m}$, against $16.1 \mu\text{m}$ for a straight line. The chain improved by a factor 4 to 13 at every length. Composite stepping went from six times worse than the single giant step to 13 percent worse. And an untrained residual network, by construction, already scores $15.5 \mu\text{m}$ against the straight line’s $16.1 \mu\text{m}$, so the trivial baseline is inside the parameterisation rather than something the network must rediscover.

What is left is one leg. The cross-magnet crossing inside a shared network sits at $1026 \mu\text{m}$ against a $46 \mu\text{m}$ ceiling on its own population, a factor 22, and the redesign moved it by only a factor 2.5. It is not width: 4×50 and 4×100 agree within ten percent on it. It is not the

output scale: leg B is the one leg where the absolute scale was roughly right, which is precisely why it had least to gain. Three further observations sharpen the target.

1. **The supervised twin is also stuck**, at $512\ \mu\text{m}$ on leg B, a factor 11 above the ceiling, while it reaches 0.06 and $0.01\ \mu\text{m}$ on the other two legs. Whatever limits leg B limits both losses, so it is not a property of the label-free objective.
2. **Leg B is a minority of the training states**. Only 1060 of the 7935 training states, 13 percent, are cross-magnet legs, and 5558, 70 percent, are plane-to-plane hops. Under the absolute parameterisation that did not matter, because the population-wide output scale was set by the long legs and they dominated the loss anyway. Under the residual parameterisation every leg is $O(1)$ by construction, so every leg now contributes comparably to the loss, and the gradient is dominated by the plane-to-plane majority. The redesign fixed the conditioning and, in doing so, handed the optimisation budget to the easy legs.
3. **The error on leg B grows monotonically along the step** in both designs, roughly a factor ten from the first node to the endpoint, which is the signature of a bend that is being under-resolved rather than of an offset.

Those three together make the cross-magnet leg inside a shared network the binding constraint on the product, and they suggest the remedy is a change to what the network is asked to carry rather than to how much of it there is.

6.3 Caveats, stated plainly

- **The tails did not improve with the medians**. On legs A and B the 95th percentile of the residual arm is no better than the absolute-output arm's, and the residual twin's tail on leg A is twice as large, Table 13. The population responsible is the soft, hard-bending tracks for which the first-order scale of Eq. (15) is a poor estimate. An extrapolator with a bounded time budget has to survive its tail, not its median, so this is a real limitation and not a presentational one.
- **One unconverged seed**. In the general-leg second wave one of the ten label-free seeds at the widest architecture stopped at 111 restarts, far short of its 400-restart cap, in the re-stalled-but-still-moving state described below. It was resubmitted but the farm queue was long, so it is reported unconverged and excluded from every median. It ranks third on the validation chain and does not affect the seed selection.
- **Ceilings are population-dependent**. The $29\ \mu\text{m}$ figure published in July was measured on 32 momentum-stratified legs of an earlier sample and is not the ceiling for any test set used here. Every ceiling in this paper is measured on the population being scored, which is why leg B's ceiling is $46\ \mu\text{m}$ in the general-leg experiment and $23\ \mu\text{m}$ on the frozen leg: the general-leg population contains softer tracks on that geometry. Mixing the two conventions would change a factor 56 into a factor 90 or a factor 36 depending on which is picked, and neither would mean anything.
- **The harness cap incident**. The shared trainer's restart cap counted the stall phase and the confirmation pass together, so runs that stalled near the cap were recorded as unconverged when they were about to converge. Three experiments hit this independently on the same day. The default has since been raised from 150 to 400; the cleaner fix, giving the confirmation pass its own small budget, is not yet applied. Nothing about the results changes, because a larger cap cannot affect a run that stopped on the stall criterion below it, so the resumed runs are what the grids would have produced under the larger cap from the start. It is recorded here because it cost a day and will cost someone else one.

- **Single-thread comparability.** The September runs are single-threaded and the July runs used four threads. A different reduction order changes the last bits, and 200 line-searched iterations amplify that into the fourth significant digit, so comparisons across that boundary are made at the level of seed ranges. The stage sweep at $q = 8$ gives 165 to 240 μm against the July baseline’s 177 to 235 μm on a dataset checked bitwise identical, which is the agreement one should expect and not better.
- **Material effects are out of scope** by construction, and gate G2 of Table 2 measures how large they are: a median of 22 μm overall, 86 μm below 2 GeV. A surrogate that reached the 23 μm ceiling would be at the level where the material the particle actually passed through starts to matter more than the propagation.

7 Outlook

7.1 The cross-magnet leg inside a shared network

The binding constraint is now specific enough to attack directly, and the diagnosis in Section 6 suggests two experiments that separate the candidate causes.

A cross-magnet-only network with variable planes. Train the residual parameterisation on leg B alone, with the start plane and the step length still inputs so that the network must serve a family of magnet crossings rather than one frozen geometry. If it recovers most of the factor 22, the problem is that leg B is a 13 percent minority competing with a 70 percent majority for the optimiser’s attention, and the remedy is a training mix. If it does not, the problem is that a 5.2 m crossing genuinely needs more than eight stages or more than a first-order scale, and the remedy is in the parameterisation or the scheme.

A stratified training mix. The same dataset with the leg types balanced rather than left at the population’s own proportions. This is the cheaper of the two and answers the same question from the other side. Both are one-line changes to the dataset builder, which is why they are the next thing to run.

Beyond those, two smaller items follow directly from the campaign. The confirmation pass should have its own restart budget rather than sharing the stall phase’s cap. And the exact-scheme solver still hard-wires the magnet-down field and had to be copied to measure the reversed-polarity ceiling; it should take the field as an argument, the way the derivative function and the reference integrator already do.

7.2 The tails

The first-order scale of Eq. (15) is right to within ten percent at the median and wrong by a large factor on the soft tracks that dominate every tail in this paper. A second-order correction to the scale, or a scale that folds in the curvature of the straight-line path through the field rather than only the field magnitude along it, is the obvious next thing to try, and it remains label-free because everything it needs is still an input.

7.3 The parked continuous-time column

The source work’s other construction takes z itself as an input and is trained on the pointwise residual of the differential equation rather than on the reconstruction residuals of a Runge–Kutta scheme. It has no tableau, no fixed set of stage states and no discretisation ceiling, but also no fixed arithmetic cost per step and a much weaker constraint per training point. The decision whether to open it was parked until the residual redesign had been measured, because the redesign addressed the specific failure that would otherwise have been the reason to switch.

The stability literature of Section 2.3 makes concrete predictions about what would happen if it were opened here, and they are worth recording before rather than after.

- **There is no isolated fixed point to park on.** The right-hand side of Eqs. (3) to (5) vanishes only where the slopes vanish and the transverse field does too, which for a track moving forward through the spectrometer is not a state it can sit at. The specific failure mode that Rohrhofer et al. [5] document, and that the regulariser of Babic et al. [7] was built to price, therefore does not have an obvious analogue here.
- **Flow-following splices should be expected instead.** The splice construction of Wang et al. [6] needs only that the spliced-in piece solves the equation near the collocation points, and the system has a five-parameter family of solutions through every plane. A candidate that follows the true track for part of the interval and a neighbouring track thereafter, with the hand-over hidden between collocation points, has the same near-zero empirical residual. That is the family the van der Pol study found the regulariser converting parking into, and it is the family pseudo-time stepping prices. So the intervention to reach for on the LHCb problem would be pseudo-time stepping with resampling, not the regulariser.
- **The difficulty axes are the bending integral and the chain.** On the oscillator the axis was the number of periods in the window. Here the natural analogue of a long horizon is a large bending integral I_B , which is what makes a leg hard, and the number of chained legs, which is what makes a path hard. A continuous-time study on this problem should sweep those two and nothing else first.

7.4 What would make the surrogate deployable

Three things, in order.

1. **The cross-magnet leg inside the shared network at the level of its own ceiling,** or a stated and understood factor above it. At $1026\,\mu\text{m}$ against $46\,\mu\text{m}$ it is not there. This is the item everything else waits on.
2. **A tail the trigger can budget for.** Median accuracy is not what a real-time system schedules against. A 95th percentile of 27 mm on the magnet crossing is not usable no matter what the median is.
3. **A measured throughput comparison against the existing extrapolator,** in the deployment precision and on the deployment hardware. The cost argument of Section 5.7 is an operation count, 44 root-finder residual evaluations against one forward pass, and an operation count is a promise rather than a measurement. Everything in this paper runs in double precision on one CPU core because that is what makes the physics comparisons clean; nothing here says what the same network does in reduced precision on a GPU, which is where it would have to live [2].

8 Conclusion

The discrete-time construction of Raissi, Perdikaris and Karniadakis [3] transfers to LHCb track extrapolation. Trained purely by the reconstruction residuals of an eight-stage Gauss–Legendre scheme, with no labels anywhere in the loss, a small network takes a single 5178 mm step across the LHCb magnet with a median endpoint error of $115\,\mu\text{m}$, three and a half orders of magnitude better than ignoring the magnet, in one forward pass against the 44 root-finder iterations the same scheme needs when solved directly. Where the scheme limits, the network reproduces the scheme’s own error to within a percent. Where the network limits, the limit is the optimiser rather than capacity, evidenced by a loss-error rank correlation of +0.965 across 96 converged runs. The method works without labels in the strongest sense available: it reaches the same

accuracy on a field polarity for which no labelled sample exists, and which is the polarity the sample itself was simulated with (Section 4.1).

The product is closer than it was but not there. A single network serving every leg type, with its outputs parameterised as the deviation from a straight line normalised by a per-leg bending integral built from the inputs alone, reaches $2.1\text{ }\mu\text{m}$ on the whole test split against $16.1\text{ }\mu\text{m}$ for a straight line, beats the trivial baseline on every leg type, and walks a real particle path four to thirteen times more accurately than the design it replaced. What it does not do is the one step the extrapolator exists for: the cross-magnet crossing inside that shared network sits a factor 22 above the scheme it is solving, its supervised twin a factor 11, and after two or three chained legs that single leg dominates everything else. That is the only failure left to find, and Section 7 says how to look for it.

Data availability

The code, the committed result tables and the figures are in the repository `LHCb.Extrapolation.Project` at <https://github.com/GeorgeWilliam1999/LHCb-Extrapolation>. Every figure in this paper is regenerated by its experiment folder's `plot.py` from committed CSV tables only. Datasets and model weights are not committed; they regenerate from the scripts, the recorded seeds and the committed metadata. Appendix A gives the commits, the cluster identifiers and the figure-to-file map.

A Provenance

Repository and commits

Repository `LHCb.Extrapolation.Project`, branch `main`. Local checkout `/data/bfys/gscriven/LHCb.Extrapolation.Project`.

- `4fe2491b4e3df024b92740b43e07265d886d758e` (5 September 2026): the four experiments of Sections 5.2 to 5.5 and the shared package. Pushed.
- `333edf7` (6 September 2026): the residual arm of Section 5.6, its results, chains and verdict. **Local, pending push.**
- `29e9928` (6 September 2026): the second wave at 4×200 in Tables 7, 8 and 9. **Local, pending push.**
- Earlier commits referenced: `38154b7` and `f31d16f` (18 July 2026, the data-generation programme and the reference card); `33cddc4` and `4c1c9f8` (21 July 2026, the first one-step network and its rerun with the fiducial requirement); `ffb7e66`, `8f82f7e` and `4447ff0` (the general-leg and residual machinery).

The experiment identifiers used in the day's plan, which appear nowhere in the body of this paper, map as follows: A1 is the stage-count sweep (`Stage_count_sweep`), A2 the network size and seed study (`Network_size_and_seed_study`), A3a the general-leg network (`General_leg_network`), A3b the chaining study (`Chained_legs`) and A4 the reversed-polarity test (`Magnet_up_field`). Earlier folders: `Baseline_data_exploration`, `Simple_first_pass`, `One_step_network` and `One_step_network.v2`.

Data set and field maps

Data set `Data_generation_exploration/Official_xdigi/training.v2/train_official.v2.npz`, metadata `train_official.v2.meta.json` (sample block corrected 5 September 2026; no data changed). Source sample: LHCb TestFileDB

entry `expected_2024_minbias_xdigi`, production 00212966, five of eight files mirrored on CVMFS (19 GB), 200 events, detector description `dddb-20231017`, conditions `sim-20231017-vc-mu100`, data type 2024. 323,533 rows, split by particle 258,857/32,389/32,287 with seed 20260718. The conditions tag is **magnet-up**, while every label in the set was computed with `field.v8r1.down.bin`; the test that settles it is `R/Magnet_tracks_dataset/results/polarity_check.json`, and the correction box in Section 4.1 states what it does and does not affect.

Field maps: `/cvmfs/lhcb.cern.ch/lib/lhcb/DBASE/FieldMap/v8r1/cdf/field.v8r1.down.bin`, md5 `af284c6954d2273c637a5e766b82b58e`, and `field.v8r1.up.bin`, md5 `9e49ddc4313b589f273540e7e0bb513b`. Grid $81 \times 81 \times 146$ at 100 mm spacing, spanning $(-4000, -4000, -500)$ to $(4000, 4000, 14000)$ mm, trilinear interpolation.

Shared module `R/_shared/`: `reference.py` (equation of motion, constants, reference integrator), `irk.py` (tableau), `field_v8r1.py` and `field_torch.py` (field and its differentiable twin), `model.py` (network and both losses), `prepare.py`, `evaluate.py`, `train.py`, `smoke_tests.py`. Gates: `vendoring_parity.py` (field values bit-identical to the origin module at archive commit `1faa97e0`, maximum difference 0.0 T) and `smoke_tests.py` (six checks, including bitwise reproduction of the July baseline dataset and first optimiser restart).

Compute

Nikhef HTCondor, all jobs single-threaded and double-precision on one core.

- Stage-count sweep: cluster **5781154** (24 runs) and **5781162** (8 resumed).
- Architecture grid: cluster **5781153** (120 runs), with continuation passes **5781167**, **5781442** and **5781445**, and **5781358** (10 supervised twin runs).
- General-leg network, first wave: cluster **5781158** (26 runs) with resubmissions **5781352**, **5781353**, **5781356**, **5781357**, **5781365**, **5781367** and **5781373**.
- General-leg network, second wave at 4×200 : cluster **5781443** (13 runs, 12 converged), with **5781566** (the resubmission of the one unconverged run).
- Reversed magnet polarity: cluster **5781156** (13 runs) with **5781163** and **5781164** (3 resumed).
- Straight-line-residual redesign: cluster **5781469** (26 runs, submitted 5 September 2026 at 21:14, all 26 confirmed) with three continuation rounds recorded in `condor/jobs_residual_round2.txt` to `round4`.

Figure-to-file map

Every figure below is copied unmodified into `figures/` from the repository path given. Paths beginning `R/` are relative to the folder `Raissi_disc_time_approach/` and the one path beginning `D/` is relative to `Data_generation_exploration/`.

The table-to-file map is in `README.md` next to this source.

References

- [1] LHCb Collaboration. The LHCb upgrade I. *Journal of Instrumentation*, 19:P05065, 2024. arXiv:2305.10515.
- [2] R. Aaij et al. Allen: A high-level trigger on GPUs for LHCb. *Computing and Software for Big Science*, 4:7, 2020. doi: 10.1007/s41781-020-00039-7.

figure	file in figures/	source in the repository
1	fig_population.png	D/Official_xdigi/figures/v1_vs_v2_population.png
2	fig_baseline.png	R/One_step_network_v2/figures/one_step_results_v2.png
3	fig_stages.png	R/Stage_count_sweep/figures/error_vs_stages.png
4	fig_architecture.png	R/Network_size_and_seed_study/figures/floor_vs_architecture.png
5	fig_loss_vs_error.png	R/Network_size_and_seed_study/figures/loss_vs_error.png
6	fig_magnet_up.png	R/Magnet_up_field/figures/magnet_up_results.png
7	fig_general_legs.png	R/General_leg_network/figures/error_by_leg_and_momentum.png
8	fig_chains.png	R/Chained_legs/figures/error_vs_chained_legs.png
9	fig_residual_scale.png	R/General_leg_network/figures/residual_scale_check.png
10	fig_residual_legs.png	R/General_leg_network/figures/error_by_leg_and_momentum_residual.png
11	fig_residual_stages.png	R/General_leg_network/figures/stage_errors_residual.png
12	fig_leg_d.png	R/Chained_legs/figures/leg_d_reproduction.png
13	fig_frozen_vs_general.png	R/General_leg_network/figures/frozen_vs_general_residual.png

- [3] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.
- [4] George William Scriven. Physics-informed neural networks on the van der pol oscillator: a measured comparison of the continuous-time and discrete-time formulations, 2026. Companion document, in preparation.
- [5] Franz M. Rohrhofer, Stefan Posch, Clemens Gößnitzer, and Bernhard C. Geiger. On the role of fixed points of dynamical systems in training physics-informed neural networks. *Transactions on Machine Learning Research*, 2023. arXiv:2203.13648.
- [6] Sifan Wang, Ali Koohy, Lu Lu, and Paris Perdikaris. When PINNs go wrong: Pseudo-time stepping against spurious solutions, 2026.
- [7] Ana Babic, Franz M. Rohrhofer, and Bernhard C. Geiger. Stabilizing PINNs: A regularization scheme for PINN training to avoid unstable fixed points of dynamical systems, 2025.
- [8] Ernst Hairer and Gerhard Wanner. *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems*. Springer, Berlin, Heidelberg, 2 edition, 1996. doi: 10.1007/978-3-642-05221-7.